BY ORDER OF THE COMMANDER
42D AIR BASE WING (AETC)

MAXWELL AFB INSTRUCTION 15-101

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Weather

WEATHER SUPPORT

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This instruction implements DAFMAN 15-129, *Air and Space Weather Operations*. It also addresses HQ USAF Program Action Directive (PAD) 97-10, *Reengineered Action for Air Force Weather*, and Air Force Policy Directive (AFPD) 15-1, *Atmospheric and Space Environmental Support*. It outlines weather service provided by 42 OSS/OSW to support the 42 ABW and all tenant organizations on Maxell AFB and Gunter Annex and establishes the responsibilities and procedures for providing and using weather services. Compliance with Attachments 2, 3, 5, 6, and 7 is mandatory IAW DAFMAN 15-129. Waivers to this publication are not authorized. Send comments and suggested improvements to this instruction on DAF Form 847, *Recommendation for Change of Publication*, through channels to 42 OSS/OSW, 220 W. Ash St., Maxwell Air Force Base, AL 36112. It applies to all agencies described herein, to include the 908th Flying Training Wing (AF Res); it does not apply to the ANG. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with AFI 33-322, *Records Management and Information Governance Program*, and disposed of in accordance with the Records Disposition Schedule located in the Air Force Records Information Management System. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Department of the Air Force.

SUMMARY OF CHANGES

This update revises MAXWELL AFB AFI 15-101 in reference to the rescinded AFI15-128 being incorporated into DAFMAN 15-129. In addition, forecasts and watches, warnings and advisories (WWA) changes in DAFMAN 15-129. Recent changes to MWP, 5-day forecast, and Heat Stress are also be included in this update. The recent mission change of the 908th Flying Training Wing (FTW) from C-130s to the Formal Training Unit for the MH-139 helicopter required significant modification to the instruction.

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Chapter 1

GENERAL INFORMATION

1.1. General.

1.1.1. 42 OSS/OSW, henceforth referred to as Maxwell Weather, is the focal point for all weather-related issues on Maxwell AFB and Gunter Annex. Maxwell Weather provides or arranges for routine and specialized weather support to the 42d Air Base Wing (ABW) and all tenant organizations on Maxell AFB and Gunter Annex in accordance with this instruction and applicable directives.

1.1.2. Changes in requirements and support capabilities will be published as changes to this instruction. An annual review of this instruction will be conducted by Maxwell Weather. Waivers are not authorized.

1.1.3. According to DAFMAN 15-129, *Air and Space Weather Operations*, paragraph 2.14 states: WF/Dets primary purpose is to integrate environmental impacts into every phase of end-user operations, from planning to execution. WFs/Dets assess the mission environment to determine environmental threats and opportunities, and where possible, identify alternatives to enable mission execution. Though each unit has unique characteristics and functions based on its parent/host unit's mission, geographic location and level of command, core roles, responsibilities, processes, and procedures are largely the same. As required, WFs/Dets will: 2.14.1. Conduct Documentation functions IAW DAFMAN 15-129 Chapter 3. 2.14.2. Conduct Weather Integration and MEFP IAW DAFMAN 15-129 Chapter 4. 2.14.3. Provide Installation and Airfield Weather Services IAW DAFMAN 15-129 Chapter 5. 2.14.4. Provide Aviation Weather Forecasting services IAW DAFMAN 15-129 Chapter 6. The 557th has designated the 26th Operational Weather Squadron (OWS) at Barksdale AFB to provide TAF production and backup support for Maxwell AFB.

1.1.4. Maxwell Weather will provide resource protection, terminal aerodrome forecasts (TAFs), meteorological watch (METWATCH) for Watches Warnings and advisories (WWAs), Pilot-to Metro Service (PMSV), flight briefing support for local and transient aircrews, and tactical-level weather products and information needed for mission execution.

1.2. Maxwell Weather Mission. Maxwell Weather is organized, trained, and equipped to conduct weather operations to provide timely, accurate, and relevant weather information and products to professionally support the 42 ABW and all tenant units on Maxwell AFB/Gunter Annex. More specifically, Maxwell Weather will take and disseminate all required surface weather observations to include all prescribed elements to local customers over the Joint Environmental Toolkit (JET); The 26 OWS produces Maxwell's Terminal Airdrome Forecast (TAF) and Maxwell Weather will back the 26 OWS in the event they are unable to produce the TAF. Maxwell Weather will issue watches, warnings, and advisories; provide Mission Weather Products (MWP) and flight weather briefings for local and transient aircrews and conduct MISSIONWATCH for these forecasts; and act as "eyes forward" for the 26 OWS, during Maxwell hours of operation.

1.3. Assumptions.

1.3.1. This instruction consolidates weather support capabilities and procedures for peacetime operations. It does not cover special or wartime operations. Procedures for wartime operations are contained in other applicable documents.

1.3.2. This instruction constitutes agreements for weather service at the local level; therefore, this instruction is published in lieu of Letters of Agreement, Memorandums of Agreement (MOA), and Memorandums of Understanding between Maxwell Weather and supported organizations.

1.4. Operational Constraints. Maxwell Weather will provide weather information based on data received from the 557th Weather Wing (557th WW), Air Education and Training Command (AETC), the 26 OWS and the National Oceanic and Atmospheric Administration (NOAA) and all sub-agencies i.e., National Weather Service (NWS), National Hurricane Center (NHC), etc. The extent of Maxwell Weather operational capabilities depends on the 557 WW, AETC, 26 OWS, and NOAA remaining operational.

1.5. Shortfalls and Limiting Factors.

1.5.1. The ability of Maxwell Weather to perform its mission is dependent on the number of personnel available and communication outages/failures. During peak hours or severe weather conditions, out of office weather briefings may not be possible due to limited manning.

1.5.2. The official point of observation for manual observations is located approximately 20 feet from the northwest corner of Building 844. The field of view from this location is significantly limited due to buildings and trees on base. Specifically, the field of view is limited to less than 1/4 mile from 080 – 170 degrees, approximately 1/4 mile from 170 – 230 degrees, 1 1/4 mile from 230 – 290 degrees, and 1/4 to 1/2 mile from 300 – 080 degrees. Only a small section of the central part of the runway is visible from 230 – 280 degrees. Also, high-intensity lights on the flight line significantly limit the ability to accurately determine visibility, cloud amounts, and cloud heights at night. The point of observation for automated observations is located at the sensor group near the approach end of runway 15. There are no limitations at this site.

1.5.3. Only the approach end of runway 15 has weather sensors. The approach end of runway 33 does not; however, data from the 15 end sensors is considered representative for non-precision approach landings on the 33 end.

1.6. Maxwell Weather Operating Hours. Weather services are available beginning at 0600 CT and continues until 2200 CT, Monday through Friday and 0900 CT to 1800 CT on weekends. Maxwell Weather closes on federal holidays, holiday weekends, and may close on Air Education and Training Command (AETC) Family Days. Staff services are available normally from 0700 CT to 1500 CT, Monday through Friday, except federal holidays, however, upon request, can be outside these hours.

1.7. Maxwell Weather Duty Priorities. The Maxwell Weather duty priorities are listed in **Table 1.2** and exist to balance limited manning and mission critical tasks. Duty priorities focus efforts

during peak work periods prone to task saturation and priority conflicts. Weather forecasters will use good judgment in complying with these duty priorities, especially where there is imminent danger to life and property.

Table 1.1. Contact Information.

Agency	Contact Information
Forecaster Phone	953-2071/3097
Fax	953-5164
Weather Manager Phone	953-3868
Assistant Weather Manager Phone	953-2071
Pilot-to-Metro-Service Radio (UHF)	342.3

Table 1.2. Maxwell Weather Duty Priority Listing.

Order Of Priority	Duties
1	Perform emergency war order taskings
2	Execute weather station evacuation
3	Respond to aircraft/ground emergencies
4	Respond to Pilot to Metro Service (PMSV) Contacts
5	Augment automated surface weather observations for mandatory parameters (See Note)
6	Provide Eyes Forward / Collaborate with 26 OWS in issuing weather warnings, watches, and advisories
7	Conduct SWAP Operations
8	Coordinate TAF and amendments with 26 OWS (prepare and issue TAF when necessary) (See Note)
9	Relay PIREPS to 26 OWS
10	Disseminate PIREPs (See Note)
11	Perform MISSIONWATCH
12	Provide Local Weather Briefs for flying units
13	Provide other briefings and forecasts
NOTE: During JET outages, the forecaster will call local users to pass observations, TAF, PIREPs, etc., before calling the 26 OWS for longline transmissions.	

1.8. Basic Weather Watch. The FMQ-19 Automated Meteorological Station is normally operated in automated mode. When in this mode, it provides a continuous weather watch for the airfield. When the FMQ-19 is being augmented, the weather forecaster will conduct a Basic Weather Watch for the airfield.

1.8.1. Basic Weather Watch (BWW) is a program that has been implemented to establish the minimum requirements needed to ensure the proper level of weather watch is maintained.

1.8.2. Because of other weather duties, weather forecasters cannot monitor the weather continuously. Due to these other weather duties, weather forecasters may not detect and report all weather changes as they occur. As a result, Maxwell Weather will conduct a BWW when the FMQ-19 is being augmented.

1.8.3. During a BWW, weather forecasters will recheck weather conditions, at intervals not to exceed 20 minutes since the last observation/recheck, to determine the need for an Aviation Selected Special Weather Report (SPECI), when any of the following conditions are observed to be occurring or are forecast to occur within 1 hour:

1.8.3.1. Ceiling forms below or decreases to less than 2,000 feet.

1.8.3.2. Ceiling dissipates or increases to equal or exceed 2,000 feet.

1.8.3.3. Visibility decreases to less than 3 statute miles (4800 meters).

1.8.3.4. Visibility increases to equal or exceed 3 statute miles (4800 meters).

1.8.3.5. Precipitation (any form).

1.8.3.6. Thunderstorms

1.8.3.7. Fog or Mist.

1.8.3.8. All supplemental criteria specified in Table 2.1.

1.8.3.9. During mandatory back-up specified in Table 2.1.

1.8.3.10. In addition to the above minimum requirements, weather technicians will remain alert for any other changes in weather conditions that will require a SPECI observation. Weather technicians will also monitor local area observational and forecast products as often as necessary to keep abreast of changes expected to affect their area of responsibility.

1.9. Cooperative Weather Watch.

1.9.1. To ensure Maxwell Weather disseminates the most accurate surface weather observations, the control tower and weather personnel will employ a Cooperative Weather Watch.

1.9.2. Of primary concern is the report of tower visibility different from the prevailing surface visibility, reporting of sector visibility, local pilot weather reports (PIREP), and any occurrence of previously unreported weather conditions that could affect flight safety or be critical to the safety or efficiency of other local operations and resources. As a minimum, Air Traffic Control (ATC) and Maxwell Weather will accomplish the following actions:

1.9.2.1. ATC will evaluate tower visibility to report changes in tower prevailing visibility and sector visibility to Maxwell Weather when tower visibility is less than 4 statute miles and is different from the surface prevailing visibility and reported sector visibility.

1.9.2.2. Weather forecasters will reevaluate weather conditions whenever a reliable source (i.e., ATC, pilots, local law enforcement, etc.) reports weather conditions different from the last disseminated observation (e.g., different ceiling height, visibility, present weather).

Based on reevaluation of the different weather conditions reported, weather forecasters will:

1.9.2.2.1. Generate a special observation (SPECI) if the different conditions warrant immediate dissemination.

1.9.2.2.2. Include the differing conditions in the next required Meteorological Aviation Report (METAR) or SPECI observation if the conditions alone do not warrant immediate dissemination.

1.9.3. ATC Personnel Actions. ATC directives (i.e., AFMAN 13-204v3, *Air Traffic Control*; FAA Order 7110.65, *Air Traffic Control*) require certified control tower personnel to make tower prevailing and sector visibility observations when the prevailing visibility at the usual point of observation, or at the tower level, is less than 4 miles. Control tower personnel certified to take visibility observations will do the following:

1.9.3.1. Notify the weather unit when they observe tower prevailing visibility to decrease to less than or increase to equal or exceed 4 miles.

1.9.3.2. When the prevailing visibility at the tower or the surface is less than 4 miles, report all changes of one or more reportable values to the weather unit.

1.9.3.3. As required by FAA Order 7110.65, use the lower of either the tower or surface visibility as the prevailing visibility for aircraft operations.

1.9.3.4. Relay all PIREPs to the Maxwell Weather forecaster not later than 5 minutes after receipt.

1.9.4. When in manual observation mode, Maxwell Weather Personnel will:

1.9.4.1. Notify the tower as soon as possible whenever the prevailing visibility at the observation point decreases to less than or increases to equal or exceed 4 miles.

1.9.4.2. Re-evaluate surface prevailing or sector visibility, as soon as practicable, upon initial receipt of a differing control tower value, and upon receipt of subsequent reportable changes at the control tower level.

1.9.4.3. Use control tower values of prevailing or sector visibility as a guide in determining the surface visibility when the view of portions of the horizon is obstructed by buildings, aircraft, etc. The presence of a surface-based obscuration, uniformly distributed to heights above the level of the tower, is sufficient reason to consider the weather unit's prevailing visibility to be the same as the control tower level.

1.10. Disseminating Weather Products.

1.10.1. JET is the primary dissemination system for surface weather observations, terminal aerodrome forecasts, weather watches, weather warnings, weather advisories, and pilot reports

on Maxwell AFB. All agencies on Maxell AFB requiring this data on a consistent basis will request a JET user account from Maxwell Weather and monitor JET online when it is needed.

1.10.2. When JET is not working, Maxwell Weather will disseminate products via phone or email, using the following priority: Maxwell Tower, Maxwell Command Post (during flying operations), airfield management operations, and OTS/OCC.

Chapter 2

RESPONSIBILITIES

2.1. 26 OWS. 26 OWS will perform the OWS responsibilities stated by the 557th under DAFMAN 15-129. Specifically, when supporting Maxwell AFB, the 26 OWS will:

2.1.1. Backup Maxwell Weather for issuing weather products, due to outages, non-duty hours or other interrupted services where Maxwell Weather is unable to produce/send weather products.

2.1.2. Maintain responsibility for WWAs at geographically separated locations (e.g., Lake Martin, Lake Jordan).

2.1.3. Perform required hurricane/tropical storm METWATCH for Maxwell and serve as the liaison between the National Hurricane Center and the base.

2.1.4. Cross-feed applicable technical exploitation and regional training products and provide technical assistance/advice upon request. Provide technical assistance/advice to the Maxwell Weather by cross-feeding applicable technical exploitation and regional training materials maintained at the 26 OWS.

2.1.5. Disseminate 26 OWS products to base agencies via standard communication systems and AFW meteorological communication systems (e.g., JET, IWWC) if Maxwell AFB personnel cannot.

2.1.6. Provide flight weather briefings for transient aircrews departing from Maxwell AFB, upon request.

2.1.7. Provide Pilot-to-Metro-Service (PMSV) to transient aircrews via phone patches or other available means. Aircrews will be required to initiate PMSV phone patches to the 26 OWS at DSN (312) 331-2651 or toll-free 1-866-223-9328.

2.1.8. Perform a continuous METWATCH Maxwell AFB/Gunter Annex. Upon identification of significant weather in their general area (observations, PIREPs, etc.), to include radar/METSAT signatures indicating possible severe weather, immediately notify the Maxwell Weather forecaster.

2.1.9. Coordinate with Maxwell Weather to develop and maintain a regional and site-specific Forecast Reference Material (FRM) with associated forecast aids, studies, etc. for the base.

2.1.10. Respond to requests for assistance when notified of an in-flight emergency, aircraft mishap, or ground mishap at Maxwell AFB by providing required information and, if required, by performing a weather data save (compile available weather information for possible use by an Aircraft Accident Investigation Board).

2.1.11. Notify Maxwell Weather of significant operational limitations or equipment outages, including work center evacuations.

2.1.12. Coordinate backup weather operations procedures and develop instructions to ensure continuous weather products and information is provided to Maxwell AFB.

2.1.13. Provide general public weather support only when the requirement cannot be met by other sources and only after authorization from Maxwell AFB Public Affairs. Other sources are typically civilian weather information sources such as the local newspaper, internet sites, and The Weather Channel. The 26 OWS and Maxwell Weather will respond to requests for public weather support based on the current workload, available manpower, and duty priorities.

2.1.14. Coordinate, maintain, annually review and electronically host installation data page for Maxwell AFB defining specific environmental support requirements, technical data, reference material, and contact information for each organization receiving TAF and WWA support from the OWS. This installation data page will take the place of the OWS-WF MOA. Where multiple WFs exist on one installation, separate data pages will be hosted.

2.1.15. Produce forecast reviews based on objective criteria IAW DAFMAN 15-129 and make the products available for use external to the OWS.

2.1.16. Produce a Tropical Cyclone Threat Analysis Product (TCTAP) that predicts hourly sustained wind and gust forecasts when Maxwell AFB is projected to receive sustained winds greater than or equal to 35 knots due to a tropical cyclone within the next 96 hours.

2.1.17. Create and maintain a METSAT imagery reference file (MIRF) and a radar imagery reference file (RIRF) containing location and region-specific imagery as well as general synoptic scale imagery.

2.1.18. Develop, coordinate, formally document and exercise not less than annually a Continuity of Operations Plan (COOP) to continue providing mission-essential services during significant outages and evacuation of the OWS facilities.

2.1.19. Provide weather services when Maxwell Weather is closed. Weather services can be obtained by contacting the 26 OWS at DSN (312) 331-2651 (Flight Weather Briefings) or DSN (312) 331-2627 (Weather Forecaster). The Maxwell Weather standby forecaster can be contacted at any time by calling the Maxwell Command Post and requesting a phone patch.

2.1.20. Assume responsibilities when Maxwell Weather evacuates. Maxwell Weather will relocate to the Alternate Operating Location (AOL) located in Maxwell Tower. The 26 OWS will assume responsibilities during the interim period until Maxwell Weather can resume operations. There are no limitations at this location.

2.1.21. 26 OWS Interruptions of Service.

2.1.21.1. Interruptions Less than 72 Hours. When the interruption in weather operations at the 26 OWS is expected to be 72 hours or less, responsibility for local TAF, weather WWAs, and transient flight briefings will transfer to Maxwell Weather. The 26 OWS graphics production/amendment capabilities will be provided by another CONUS OWS.

2.1.21.2. Interruptions More than 72 Hours. For an outage lasting more than 72 hours, the 26 OWS will deploy personnel to AFWA (or another OWS) and reclaim the original support provided to Maxwell Weather from a backup location.

2.2. Maxwell Weather. Maxwell Weather will:

2.2.1. Coordinate requests for briefing support with the 26 OWS on larger operations when Maxwell Weather cannot provide exploitation products to Maxwell AFB due to manning levels.

2.2.2. Provide flight weather briefings to Civil Air Patrol United States Air Force (CAPUSAF) members, during normal duty hours, performing official Air Force operational duties. CAPUSAF members performing official flight duties outside of normal operating hours should be advised of remote briefing capabilities (i.e., through the 26 OWS). Maxwell Weather will not have personnel on duty solely to provide briefings for Civil Air Patrol activities.

2.2.3. Provide service IAW posted duty priorities for walk-in requests from transient aircrews. When transient or staged aircrews request briefing support, Maxwell Weather will:

2.2.3.1. Provide a briefing or update an existing briefing form as time and resources allow. EXCEPTION: Refer flight weather briefing requests for AMC, USAFE, and PACAF integrated flight-management (IFM) missions to the appropriate IFM weather support agency (e.g., 618 AOC (TACC)/XOW, 21 OWS, and 17 OWS respectively). Maxwell Weather may provide access to meteorological satellite imagery, take off data, and other perishable weather data for IFM crews upon request, but will refer the aircrew to their supporting IFM weather organization for weather updates to the actual mission package.

2.2.3.2. If unable to provide a briefing or update an existing briefing, provide contact information for the 26 OWS and direct the aircrew to Airfield Operations-provided terminals for online access to weather support. Assist aircrew as time permits.

2.2.4. Relay pertinent information to the 26 OWS concerning changing local conditions that significantly diverge from forecasted conditions, especially affecting resource protection or the TAF.

2.2.5. Incorporate the MIRF and RIRF managed by the 26 OWS into qualification and continuation training. Maxwell Weather will contribute pertinent METSAT imagery or radar signatures that may be of training value to the 26 OWS for consideration for inclusion into the MIRF/RIRF.

2.2.6. Identify local or fine scale influences on weather parameters yielding empirical evidence that may support establishment of a rule of thumb (ROT) for Maxwell AFB. These will be identified as observed and passed on to the 26 OWS for validation. The Maxwell Weather role will not exceed gathering and furnishing of information.

2.2.7. Will forward newly created or updated documents to the 26 OWS to ensure the most current forecast reference materials (FRM) are on file.

2.2.8. Forward Rules of Thumb (ROTs) to the 26 OWS for validation and documentation of forecast effectiveness. ROTs under development will be designated as experimental‖ in local procedures and OWS-maintained FRMs until validated.

2.2.9. Review installation data pages within 90 days of assignment or annually, whichever occurs first; to ensure consistency with supported unit requirements. Updated/reviewed installation data pages will have the date of review or updated publication date published on the document. Maxwell Weather leadership will inform the 26 OWS of any changes that occur on the installation to update the installation data page. Any support issue between the 26 OWS and Maxwell Weather that cannot be resolved at the appropriate subordinate levels through the chain of command will be elevated to the functional higher headquarters for resolution.

2.2.10. Forward updated changes in documented support to the 26 OWS, as required. If these updates drive further changes to Maxwell Weather-26 OWS operations, prior coordination between Maxwell Weather-26 OWS leadership is mandatory. Agreed upon changes will be documented on the installation data page.

2.2.11. Notify the 26 OWS Operations Superintendent or regional zone OIC when temporary changes to duty hours occur (e.g., Family Day, Goal Day, other closures/extensions). This notification will be made no later than 24 hours in advance of the closure to allow the 26 OWS to reallocate its resources, as required.

2.2.12. Review the FRM (Forecast Reference Material) for the installation hosted electronically by the 26 OWS at least annually. Provide updates (e.g., new airfield weather sensors, local mission types) to allow the 26 OWS to update the installation's electronic FRM, as required.

2.2.13. Notify the 26 OWS when moving to the AOL. Provide the 26 OWS temporary telephone numbers and any changes in the weather warning and advisory notification procedures.

2.2.14. Limit analysis efforts to mission-scale effects along mission routes and operating areas (e.g., terrain influences) to refine characterization products provided by the 26 OWS.

2.2.14.1. Use 26 OWS-issued graphical and alphanumeric analysis and forecast products, TAFs, Watches, Warnings (WW), and Forecast Weather Advisories (FWA) in developing the MWP, as well as for updating supported unit(s) decision-makers on environmental impacts to operations.

2.2.14.2. Alert the 26 OWS to developing situations not coded in meteorological reports that potentially drive amendments to forecast products from the 26 OWS or impact flight safety.

2.2.15. Ensure operating procedures include timeline requirements (e.g., Air Traffic Control/Supervisors of Flying will relay pilot report information to weather not later than 5 minutes after receipt) and requirements for relaying all PIREPS to the 26 OWS.

2.2.16. Fully utilize and not deviate from the tropical cyclone information provided by the 26 OWS (i.e., Tropical Cyclone-Threat Analysis Product (TC-TAP)) derived from National Hurricane Center products.

2.2.17. Upon request from Disaster Preparedness or any other agency, obtain/provide CDMs from the 26 OWS.

2.2.18. Coordinate weather watch and warning criteria, as well as tropical cyclone forecasts, to include the Desired Lead Times and notification methods with the 26 OWS.

2.2.19. Issue weather warnings when imminent weather conditions pose a hazard to life and property. Update the 26 OWS as soon as possible.

2.2.20. Coordinate forecast weather advisory requirements with the 26 OWS and document the requirement on the installation data page. The desired lead time based upon Maxwell AFB requirements and the 26 OWS capability to provide such advance notice.

2.2.21. Develop procedures to provide appropriate information to the Maxwell Command Post IAW AFMAN 10-206, *Operational Reporting (OPREP)*, and Maxwell AFB policies. Maxwell Weather will ensure the 26 OWS and AETC Weather are aware of the reported weather event as soon as practical.

2.2.22. Relay significant, time-sensitive meteorological information not found in coded meteorological reports to the 26 OWS to assist in characterization operations as part of the Eyes Forward function.

2.2.23. Notify the 26 OWS Operations Floor Production Supervisor of all aircraft mishaps as soon as possible after notification of the event. Note: Expedient notification is critical to archiving data pertinent to the mishap.

2.3. The 26 OWS and Maxwell Weather together will:

2.3.1. Ensure the timely issuance of WWAs with supported customers.

2.3.2. Establish procedures to manage severe weather threats, to include recalling of weather personnel.

2.3.3. Determine mandatory forecast review criteria of severe weather events as required. Maxwell Weather may request reviews for severe weather events that negatively influence the effective execution of military operations. The 26 OWS is the primary OPR for the formal review, and Maxwell Weather will provide inputs and coordination, as required.

2.3.4. Assist the installation commander and disaster preparedness personnel providing technical material and/or developing presentations to educate installation agencies on the purpose, applicability and operating procedures of the weather watch and warning system.

2.4. 908th Flying Training Wing/703d Helicopter Squadron: Will call for weather information. The 26 OWS and Maxwell Weather will provide regular weather services with forecasts by sector as well as METRO radio communications. The AM MEF will be posted to the 703 HS Microsoft Teams® page no later than 0700L with a valid period 0700-2000L. The PM MEF will be posted at 1400L with a valid period of 1600-0200L. Amendments to the MEF will be posted as necessary and Maxwell Weather will notify the 703 HS Operations Desk when such amendments occur. Weather briefs for flights taking place outside of the local training area or during periods not covered by the MEF may be requested via email or phone.

2.5. Airfield Management Operations:

2.5.1. Monitors weather on JET at all times. Reports malfunctions to Maxwell Weather and requests backup dissemination. Maxwell Weather will help troubleshoot the problem and log out the system if necessary.

2.5.2. Notifies Maxwell Weather personnel, via the base secondary crash net or hotline, of all aircraft precautionary landings, emergency landings, aircraft mishaps and ground emergencies.

2.5.3. Notifies Maxwell Weather manager of the following:

2.5.3.1. Code 3 (equivalent) or higher diverts.

2.5.3.2. Changes to airfield operating hours.

2.5.3.3. PIREPs.

2.6. Airfield Manager:

2.6.1. Ensures all airfield management operations coordinators are properly trained to read and interpret weather information disseminated over JET.

2.6.2. Ensures current Maxwell Weather hours of operation and PMSV information are published in the DoD Flight Information Publication (FLIP).

2.6.3. Provides Maxwell Weather with Defense Mapping Agency, National Imagery Mapping Agency (NIMA), Flight Information Publications (FLIP), and other FAA publications as required. Maxwell Weather will review FLIPS, including the RADAR Instrument Approach Minimums, for changes in airfield minima.

2.7. 42d Operations Support Flight (ATC):

2.7.1. Coordinates training with Maxwell Weather to ensure control tower personnel are properly trained to take limited weather observations in support of the cooperative weather watch program.

2.7.2. Coordinates training with Maxwell Weather to ensure control tower personnel are trained on reading and interpreting weather information.

2.7.3. Provides indoctrination training to Maxwell Weather personnel on Air Traffic Control facilities, effect of weather on handling air traffic and the use of weather information.

2.7.4. Informs the duty forecaster of the active runway upon opening the control tower and whenever changes in runway status occur.

2.7.5. Relays all PIREPs to Maxwell Weather forecaster within 5 minutes of receipt.

2.7.6. Assists with the cooperative weather watch.

2.8. 42d Security Forces Squadron:

2.8.1. Notifies the following agencies of severe weather warnings:

2.8.1.1. All posts and patrols.

2.8.1.2. Base housing residents (when directed by Maxwell Command Post).

2.8.1.3. Family campground residents (when directed by Maxwell Command Post).

2.8.2. As part of the Cooperative Weather Watch, notifies Maxwell Weather if any significant weather (tornadic activity, damaging winds, or large hail) and/or related damage is seen on Maxwell AFB-Gunter Annex.

2.9. Maxwell Command Post:

2.9.1. Has JET operating at all times. Reports JET malfunctions to Maxwell Weather and requests backup dissemination. Maxwell Weather will help troubleshoot the problem and log out the system if necessary.

2.9.2. Ensures command post controllers can properly read and interpret data disseminated over the JET. Requests training from Maxwell Weather if necessary.

2.9.3. Immediately disseminates weather warnings over the Giant Voice (see **Attachment 6**) as soon as they are received. Weather watches and warnings are also disseminated over AtHoc.

2.9.4. After receipt of a tornado warning, will be prepared to activate the base alert siren. Activate the base siren upon notification from Maxwell Weather or 26 OWS that a tornado is expected within 15 minutes, when a tornado is occurring at Maxwell AFB, or when directed by the wing commander.

2.9.5. When the airfield is closed, notifies Maxwell Weather manager concerning any aircraft or ground mishaps.

2.9.6. Coordinate reports of weather-related damage information with WX, EOC, & CAT.

2.9.7. Maintains a current copy of Maxwell Weather recall roster and notifies Maxwell Weather manager whenever Maxwell Weather is closed and any of the conditions requiring notification/activation of SWAP standby member occurs.

2.10. 42d Communications Squadron (42 CS):

2.10.1. The server for the JET system resides within the 42 CS NCC and the 26 OWS and WF are provided with remote access. 42 CS will allow access for server TCNOs/updates or perform the install of TCNO/updates IAW 24 AF-AF/A3OW-JET PMO MOA. Contact AFWA Customer Service (Fielded Systems) at DSN (312) 271-2586, Option 2 for JET equipment issues.

2.10.2. Ensures 42 CS maintenance personnel maintain weather equipment in accordance with appropriate directives and technical orders.

2.10.3. Reports to higher headquarters information concerning outages on meteorological communications or sensing equipment that impact mission capability.

2.10.4. On-call Job Controller issues Job Control Numbers, contacts appropriate maintenance personnel for 42 CS maintained equipment, and monitors maintenance actions. Maxwell Weather personnel will provide mission impact statements when equipment outages impact mission support capability.

2.10.5. Is responsible for maintaining all meteorological equipment listed below in the priority listed. Use restoral supply priority of Force Activity Designator (FAD) IV for all equipment except for the NEXRAD, which will use a FAD II rating. Subject to safety reasons, if more than one piece of Airfield System's equipment needs repair, the 42 CS will use the priority indicated below:

2.10.5.1. Control Tower: ETVS, radios, landlines, etc.

2.10.5.2. Localizer.

2.10.5.3. Glideslope.

2.10.5.4. TACAN.

2.10.5.5. Digital Voice Recording System (DVRS) (**NOTE:** If sufficient Airfield Systems personnel are available, concurrent maintenance on both the DVRS equipment and the NAVAIDS may be accomplished).

2.10.5.6. DBRITE.

2.10.5.7. Flight Data System (FDS/FDIO)

2.10.5.8. Digital Weather Information/Wind Sensors.

2.10.5.9. Light guns.

2.10.6. 42 CS maintenance personnel will be available 24 hours a day/7 days a week and will have a 1 hour response time.

2.10.7. 42 CS maintenance personnel will notify Maxwell Weather one hour prior to taking equipment offline to work on it.

2.11. 42 ABW Chief of Safety: Ensures the convening authority includes a qualified weather representative as a member of any Interim Safety Board or Safety Investigation Board whenever weather or weather service may be a contributing factor to the accident under investigation.

2.12. Base Logistics: Ensures all weather watches and warnings are forwarded to the Fuels Control Center and Vehicle Maintenance.

2.13. Civil Air Patrol – U.S. Air Force (CAP – USAF): Notifies Maxwell Weather manager of any changes to aircraft weather sensitivities listed in **Attachment 5.**

Chapter 3

SERVICES PROVIDED

3.1. Surface Weather Observation.

3.1.1. All surface weather observations will be disseminated over the Joint Environmental Toolkit (JET) and will include the time of the observation, wind direction and speed, visibility, present weather, sky condition, temperature in Celsius, dew point in Celsius, altimeter setting, and remarks. An instantaneous reading of the Runway Visual Range (RVR) will be included when visibility is one mile or less or RVR is 6,000 feet or less. Surface observation format is listed in **Attachment 4**.

3.1.2. Special observations will be disseminated whenever criteria listed in **Attachment 2** occur.

3.1.3. FMQ-19 Augmentation. The FMQ-19 will normally be operated in auto mode. However, there are times when data from the FMQ-19 will need to be augmented. Augmentation is the process of manually adding or editing data in an observation generated by the FMQ-19. The two augmentation processes used are supplementing and back-up. Supplementing is a method of manually adding meteorological information to an automated observation that is beyond the capabilities of the FMQ-19 to detect and/or report. Back-up is the method of manually providing meteorological data and/or dissemination to the FMQ-19 observation when the primary automated method is not operational or unavailable due to sensor and/or communication failure.

3.1.3.1. Maxwell Weather will use manual observing methods (e.g., prevailing visibility reporting, thunderstorm reporting) when supplementing or backing up the FMQ-19.

3.1.3.2. Any data used to supplement or back up an observation will have been evaluated within 15 minutes of the actual time of the report. Data older than will 15 minutes will not be used in an observation.

3.1.3.3. In order to properly supplement and back up the FMQ-19, the weather forecaster on duty will maintain situational awareness of current weather conditions and the FMQ-19 observations at all times.

3.1.3.4. Supplementing. The weather forecaster on duty will supplement observations when the airfield is open, and the weather conditions are observed to be missing or incorrect. Also, the forecaster will perform a Basic Weather Watch from the manual observing point of observation and be ready to supplement/back-up observations if the conditions in Table 3.1 are forecast to occur within 1 hour.

3.1.3.5. Back-up. Except for some automated remarks, back-up refers to the weather forecaster providing the same reporting capability as that provided by the FMQ-19. The on-duty forecaster will back-up the FMQ-19 if the system/sensor(s) is/are not operational or unavailable due to sensor and/or communication failure. There is no requirement to back-up the system/sensor when the airfield is closed unless tornadic activity is occurring

or forecast to occur. Weather technicians will provide back-up information in accordance with Table 3.2, Summary of Back-up Encoding, and Table 3.3, Summary of Back-up Remarks, in AFMAN 15-111, *Surface Weather Observations*.

3.1.3.6. When using backup wind and pressure equipment, values will be estimated.

Table 3.1. Mandatory Supplementary/Back-Up Weather Conditions.

Mandatory Supplementary Weather Conditions – Body of Report
1. Tornado (+FC)
2. Funnel Cloud (FC)
3. Hail (GR) ≥ ½ inch
4. Volcanic Ash (VA)
5. Ice Pellets (PL)
Mandatory Supplementary Weather Conditions – Remarks Section
1. Tornadic Activity
2. Snow depth (only during airfield operating hours)
Mandatory Back-Up Weather Conditions
1. Thunderstorms (within 5NM of Maxwell AFB)[4]
2. Ceiling/Visibility below 1500ft/3 miles[4]
3. Freezing precipitation, any type[5]
4. Snow[5]
5. Freezing fog[5]
NOTES: 1. The immediate reporting of +FC or FC takes precedent over any other phenomena. 2. The forecaster will log on to the FMQ-19 and be prepared to supplement for tornadic activity anytime a weather watch or warning is issued for tornadoes. 3. Data augmented in an observation must have been evaluated within 15 minutes of the actual time of the report. 4. At times the FMQ-19 automated observations have been unrepresentative when these conditions occur. These unrepresentative observations have caused mission delays and cancelations when aircraft were doing visual approaches and assault strip landings. Indicated items will be backed up when flying operations are occurring. 5. Due to a deficiency in the FMQ-19 - incorrectly reporting freezing precipitation during snow or freezing fog events.

3.2. Terminal Aerodrome Forecast (TAF).

3.2.1. TAFs provide meteorological information in a standard text format containing the cloud cover, cloud heights, and visibility for general flight rule conditions, as well as wind, altimeter, and other weather parameters needed to sustain the landing and takeoff of aircraft.

3.2.2. 26 OWS issues the official TAF for Maxwell AFB. Maxwell forecasters will issue TAFs as a backup for when the 26 OWS is unable to prepare one.

3.2.3. TAFs will be issued by 0615L 1415L and, if needed 2215L, when the airfield is open, and will be valid for a 30-hour period. See **Attachment 4** for code breakdown.

3.2.4. Unless otherwise specified, forecast elements in the main body of the forecast text apply to the area within a 5 nautical mile radius of the center of Maxwell airfield complex.

3.2.5. 26 OWS and Maxwell Weather will ensure forecasts indicate when conditions cross specification criteria, are representative of current and expected weather conditions, and remain within standard amendment criteria. TAFs will be amended and disseminated whenever thresholds in **Attachment 3** are crossed and are not reflected in the current TAF. Amendments will cover the remaining time in the original 30-hour TAF.

3.2.6. The following remark will be appended to the 1400L TAF, unless the 2200 TAF is needed, IAW AFMAN 15-124, *Meteorological Codes*, "LAST NO AMDS AFT YYGG NEXT YYGG" where AFT XXGG is the day and time (Z) when the airfield is closed and a TAF is no longer required and NEXT YYGG is the time when the new TAF is required. For example, "LAST NO AMDS AFT 2004 NEXT 2212" means the TAF will not be amended after 04Z on the 20th and the next TAF is scheduled for 12Z on the 22nd day of the month.

3.3. 703d Helicopter Squadron Mission Weather Product (MWP). A MWP is coordinated and adjusted for changes as necessary, in coordination with the 703 HS for mission changes. Two Mission Execution Forecasts (MEF) will be issued to cover daily flights. The AM MEF will be posted to the 703 HS Microsoft Teams® page no later than 0700L with a valid period 0700-2000L. The PM MEF will be posted at 1400L with a valid period of 1600-0200L. Amendments to the MEF will be posted as necessary and Maxwell Weather will notify the 703 HS Operations Desk when such amendments occur. Weather briefs for flights taking place outside of the local training area or during periods not covered by the MEF may be requested via email or phone.

3.4. Flight Weather Briefings.

3.4.1. Flight weather briefings will be prepared using a DD Form 175-1, *Flight Weather Briefing*. Pilots may request verbal briefings instead of a flight weather briefing form. Verbal briefings will be logged on a Maxwell AFB Form 27, *Aircrew Briefing Log*. Information on the DD Form 175-1 can be found in Attachment 8 of DAFMAN 15-129.

3.4.2. Maxwell Weather will provide flight weather briefings to aircrews assigned to Maxwell AFB, i.e., the 703d HS and CAP-USAF. For Maxwell aircrews that are off-station, Maxwell Weather will make arrangements to provide briefings remotely (i.e., via phone, fax, web, or email) or arrange with other Air Force weather units or Operational Weather Squadrons to provide the crews with weather support.

3.4.3. Maxwell Weather will provide flight weather briefings to transient aircrews as local mission and manning dictate. Transient aircrews will be directed to the 26 OWS when Maxwell Weather is unable to provide MWPs due to time constraints.

3.4.4. 26 OWS will employ a web-based capability to provide remote flight weather briefings. Sections of the web page will be dedicated to presenting weather products for aircrews during a telephone-based briefing with visual aids designed to enhance briefing effectiveness. The web page will also allow aircrews to make on-line, advanced requests for briefings.

3.4.5. To ensure 26 OWS has adequate time to examine the weather conditions and complete the required documentation, aircrews will need to request routine briefings at least 2 hours prior to brief time. Aircrews are encouraged to request routine briefings via the 26 OWS web page. If the web page is not working, briefings can be requested via telephone at DSN (312) 331-2651 or commercial 318-529-2651 (ONLY if DSN is not available), toll free 1-866-223-9328 or via fax at DSN (312) 331-2609. **Note:** The toll-free number above is reserved for aircrews at civilian facilities without access to DSN phone communications.

3.4.6. No-notice and short-notice requests (less than 2-hour notice) for flight weather briefings will be accomplished as quickly as possible depending on the current workload, available manpower, and duty priorities. No-notice briefing requests will be prioritized behind existing requests unless special circumstances warrant a higher priority (e.g., alert, search and rescue, medical evacuation, distinguished visitor.)

3.4.7. 26 OWS/Maxwell WF will document routine flight weather briefings using the appropriate briefing form (e.g., DD Form 175-1, *Flight Weather Briefing*) or electronic equivalent. The briefing form will be forwarded to the aircrew by whatever means is most appropriate and feasible (e.g., fax, e-mail, 26 OWS web page).

3.4.8. 26 OWS/Maxwell WF will document verbal flight weather briefings using the 26 OWS Verbal Flight Weather Briefing Log.

3.4.9. MISSIONWATCH will be performed for all MWPs and flight weather briefings provided by Maxwell Weather. Whenever unforecasted changes occur, or are expected to occur, Maxwell Weather will notify the pilot through the 703 HS Duty Officer/NCO, ATC, airfield management operations, or, when the aircraft is within 15 miles of Maxwell AFB, notify the Maxwell Command Post if possible.

3.5. Pilot-to-Metro-Service (PMSV).

3.5.1. Maxwell Weather provides PMSV radio support during hours of operation using frequency 342.3 MHz. There are no known limitations for this radio/frequency.

3.5.2. PMSV support is not available locally during times when Maxwell Weather is closed. However, the 26 OWS provides support by phone patch 24 hours a day/7 days a week at DSN (312) 331-2651/2652/2653.

3.5.3. Information from PMSV contacts will be logged on a Maxwell AFB Form 28, *Pilot – to-Metro-Service Log*. See **Figure 3.1** for an example.

Figure 3.1. Maxwell AFB Form 28, *Pilot-to-Metro-Service Log*, Example.

PILOT TO METRO SERVICE (PMSV) LOG					MONTH Oct 04	
DAY	TIME (Z)	CALL SIGN	INFORMATION GIVEN PILOT/COMMENTS	PIREPS	INIT	DISSEM
12	1406	TOIL60	KMXF FT 1700Z	**OV/ KMXF36010 /TM 1406 /FL 120 /TP C130** /SK BKN100-TOPUNKN /WX FV99 /TA M08 /WV 26045 /TB LGT CHOP /IC NEG /RM	MSG	X
14	1826	JIFFY29	KBHM	**OV/ KBHM /TM 1826 /FL 300 /TP B707** /SK /WX /TA M28 /WV 28075 /TB LGT 250-300 /IC NEG /RM	WC	X
				OV/ /TM /FL /TP /SK /WX /TA /WV /TB		

3.5.4. Outage procedures.

3.5.4.1. If the PMSV is inoperative, the tower (ATC) will be notified and they will monitor the frequency, and the radio will be logged out with job control.

3.5.4.2. For long-term outages (more than 1 hour), a NOTAM will be transmitted with information about the outage including the expected length, if known, i.e., "PMSV 342.3 unavailable until approximately 24/1800Z. Tower monitoring frequency 342.3."

3.5.4.3. When the equipment becomes operational, the tower will be notified, and the radio will be logged back in with job control. Also, if a NOTAM was transmitted reporting the PMSV as being down, base operations will transmit a NOTAM reporting that the PMSV frequency is back in operation.

3.6. Pilot Reports (PIREP).

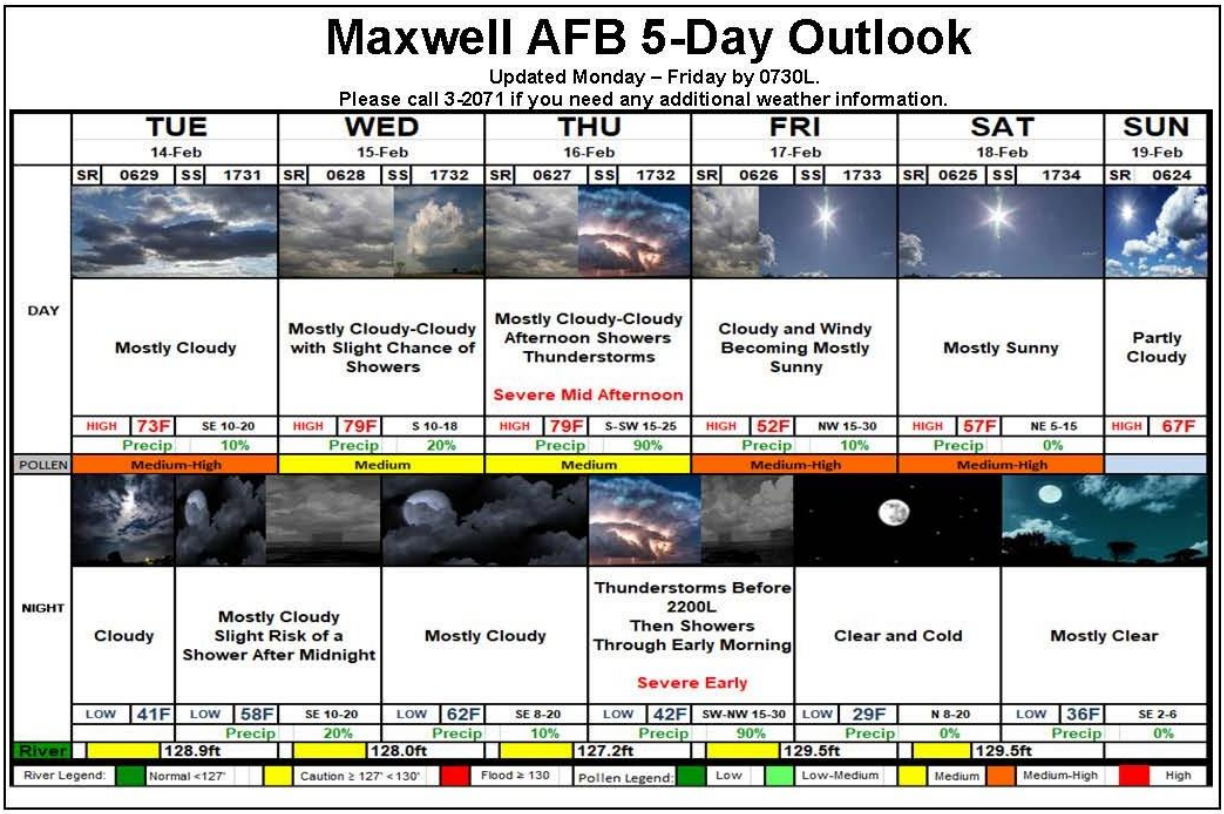
3.6.1. Aircrews are highly encouraged to provide PIREPs during PMSV contacts.

3.6.2. All PIREPS will be disseminated over the JET. See **Attachment 4** for format.

3.6.3. Maxwell Tower will relay all PIREPS they receive to Maxwell Weather within 5 minutes of receipt.

3.7. Maxwell AFB 5-Day Weather Outlooks. The Maxwell AFB 5-Day Weather Outlook will be produced and emailed to Base Populus by 0730L, Monday through Friday, excluding U.S. Holidays and wing down days. See **Figure 3.2** for an example.

Figure 3.2. Maxwell AFB 5-Day Weather Outlook Example.



3.8. Space Weather Support. At this time, no agency on Maxell AFB requires space weather on a recurring basis. However, space weather data, including solar impacts on radio communications and GPS errors, will be made available upon request.

3.9. Base Staff Meetings. Maxwell Weather will present a weather briefing at 42 ABW Base Staff meetings upon request. The content of the briefing is flexible, but will include, as a minimum, 5-day forecast, and any significant event expected for Maxwell AFB.

3.10. Emergency Operations Center (EOC) Briefings. Maxwell Weather will provide a weather representative when the EOC stands up. When a weather briefing is required, the EOC support team will notify Maxwell Weather as soon as possible with the briefing time, briefing location, and weather information needed.

3.11. Weather Training.

3.11.1. Air Traffic Control Training/Certification. Maxwell Weather will provide initial weather training and certification for ATC personnel on METAR code, TAF code, visibility determination, and the Cooperative Weather Watch. The NCOIC of ATC Training and Standardization will provide recurring training and certification.

3.11.2. Upon request, Maxwell Weather will provide Instrument Refresher Course briefings to flying units located on Maxwell AFB.

3.11.3. Upon request, Maxwell Weather will train other non-weather personnel assigned to agencies (airfield management operations and Maxwell Command Post) that receive and take action based on weather information transmitted over the JET. Generally, this training concerns interpreting locally disseminated weather information such as observations, forecasts, weather advisories, weather watches, and weather warnings.

3.11.4. Maxwell Weather will train other personnel involved in aerodrome operations (e.g., 703 HS Duty Officer/NCO) as necessary.

3.12. Climatology.

3.12.1. Each month, Maxwell Weather will prepare a climatological data summary for Maxwell AFB on an Excel® spreadsheet, post it on their web site, and email it to organizations that require it.

3.12.2. Units may request climatological information for Maxwell AFB and other areas of interest. Data can be compiled for deployments, construction projects, runway repair/cleaning, heating and cooling-degree days, and other uses. Requests should be made with as much lead-time as possible so Maxwell Weather can compile data or obtain it from the 14th Weather Squadron. Maxwell Weather may be consulted on uses and benefits of climatological data.

3.13. Major Accident/Natural Disaster/Nuclear Fallout Support. Under normal operating conditions, Maxwell Weather will assist base agencies responding to major accidents and natural disasters. During communication outages, equipment outages, personnel shortfalls, and when Maxwell Weather is evacuated, the level of support may be degraded.

3.13.1. Major Accident Support. This category includes all aircraft accidents and ground mishaps. Weather data and forecasts will be provided to the EOC, and other base agencies as requested.

3.13.2. Natural Disasters. Maxwell Weather will provide weather support for natural disasters IAW procedures outlined in 42 ABW Plan 10-2, *Comprehensive Emergency Management Plan (CEMP)*, and the Quick Reaction Checklists (QRC). Maxwell Weather will be the single point of contact for coordinating weather information from the National Weather Service and National Hurricane Center.

3.13.3. Hurricanes.

3.13.3.1. 26 OWS will use the tracks and wind forecasts from the National Hurricane Center (NHC) tropical cyclone bulletins when developing the TAFs, warnings, watches, and advisories for Maxwell AFB. No deviation from these official forecast tracks is authorized, however, modifications to the local wind forecast may be necessary due to terrain effects.

3.13.3.2. 26 OWS will serve as the liaison between the NHC and Maxwell Weather.

3.13.3.3. Maxwell Weather will translate the official tropical cyclone forecast and 26 OWS forecasts into a specific mission forecast for Maxwell AFB.

3.13.3.4. When tropical cyclones are forecasted to affect the southeastern United States or the Gulf Coast, Maxwell Weather will notify the Maxwell Command Post.

3.13.3.5. If the EOC forms, Maxwell Weather will provide briefings that include, as a minimum, the storm's current and forecast positions.

3.13.3.6. If winds of 50 knots or greater are possible, associated with a hurricane or tropical storm, and are expected at Maxwell AFB, Maxwell Weather will assist the wing commander in determining times to go into Hurricane Conditions (HURCON) and Recovery Phase.

3.13.3.7. 26 OWS and Maxwell Weather will monitor weather conditions and issue watches, warnings, and advisories for other hurricane related hazards if they are occurring or expected to occur including tornados, strong winds, thunderstorms and lightning, cross winds, and heavy precipitation.

Table 3.2. Hurricane Conditions (HURCON)/Response Phases

HURCON 4 Destructive winds of 50 knots or greater are possible within 72.
HURCON 3 Destructive winds of 50 knots or greater are possible within 48 hours.
HURCON 2 Destructive winds of 50 knots or greater are possible within 24 hours.
HURCON 1 Destructive winds of 50 knots or greater are possible within 12 hours.
Recovery Phase Actions taken to reestablish primary mission capability and return Maxwell Air Force Base to normal operations.

3.14. Flag Conditions and Wet Bulb Global Temperature. Bioenvironmental Engineering (42 OMRS/SGXB) will provide flag conditions and wet bulb global temperature (WBGT) for Maxwell AFB. 42 OMRS/SGXB will send readings directly to Maxwell AFB Command Post. Flag conditions will be disseminated by the Maxwell Command Post over AtHOC. Maxwell Weather will produce a 5 day Forecast Heat Index product to be sent out with the 5 day, Monday–Friday. Begin by May 31st and end by October 1st.

3.15. Pre-deployment Briefings and Concept Briefings. Upon request from the 42d Logistics Readiness Squadron, Maxwell Weather will provide pre-deployment briefings, concept briefings, and climatological data for deployment locations.

3.16. Releasing Weather Information to Non-DoD Agencies. Maxwell Weather may support nonmilitary agencies if base Public Affairs has given permission. Such requests range from written requests for the release of local weather information to immediate support required to avert or lessen the loss of life, personal injury, or property damage. Support provided will not impair the military mission of Maxwell Weather. The Air Force will be reimbursed for such services, and will not assume any legal, financial, or moral responsibility for the service provided.

3.17. Chemical, Biological, Radiological, Nuclear, and High-Yield Explosive (CBRNE) Operations. Maxwell Weather will serve as the weather subject matter expert to CBRNE operations on Maxwell AFB/Gunter Annex. However, Maxwell CBRNE program is run by the Maxwell Fire Department. Maxwell Weather will:

3.17.1. Routinely meet with installation EM, Fire Emergency Services (FES), and Bioenvironmental Engineering (BEE) to achieve appropriate mission immersion.

3.17.2. Become familiar with the CBRNE plume models used on Maxwell AFB (e.g., Joint Effects Model (JEM), and Area Locations of Hazardous Atmospheres (ALOHA), etc.), their tactical decision aid outputs, and uses in installation commanders' decision cycles.

3.17.3. Understand the variety of possible weather input options within each model for each type of C, B, R, N and E event.

3.17.4. Recommend and provide the most appropriate weather data type to use to run their chosen CBRNE model to assess a real-time event.

3.17.5. Provide a model data recommendation consistent with available models at the 26 OWS.

3.17.6. If surface observations or alphanumeric forecasts are requested, make sure that observations and forecasts provided are representative of the location/time of the CBRNE event.

3.17.7. Work closely with EM or other ESF functions to ensure the wing commander gets a consistent picture.

3.17.8. Upon request, obtain/provide Chemical Downwind Messages (CDM) for Maxwell AFB.

Chapter 4

WEATHER WARNINGS, WATCHES, AND ADVISORIES

4.1. General.

4.1.1. The term meteorological watch (METWATCH) is used to describe the process of monitoring observed and forecasted weather conditions and notifying selected agencies when pre-established weather conditions occur or are expected to occur.

4.1.2. The 26 OWS, in collaboration with Maxwell Weather, will perform METWATCH duties for Maxwell AFB.

4.1.3. Warnings and advisories will be issued as either forecasted or observed. Forecasted products will have a valid time. Observed products will be valid from the time issued until further notice, i.e., until the phenomena are no longer occurring.

4.1.4. Maxwell Weather will issue all METWATCH products when they are on-duty. The 26 OWS will issue METWATCH products when Maxwell Weather is closed.

4.1.5. Maxwell Weather will be responsible for local dissemination and contact the 26 OWS as soon as possible afterward.

4.1.6. Desired lead time is the minimum amount of advanced notice an agency requires in order to take specific actions prior to the onset of a particular weather phenomenon.

4.2. Definitions

4.2.1. Weather Watch. A weather watch is a special notice to notify operational commanders of a potential for environmental conditions of such intensity as to pose a hazard to life or property. Weather Watches indicate a potential for environmental threats and are used by operational commanders to make force protection and risk management decisions.

4.2.2. Weather Warning. A weather warning is a special notice to notify operational commanders when an established weather condition of such intensity as to pose a hazard to life or property is occurring or is expected to occur. Weather warnings provide concise information outlining environmental threats and are used by operational commanders to make force protection decisions.

4.2.3. Weather Advisory. A weather advisory is a special product notifying an end user when an established environmental condition affecting operations is occurring or is expected to occur.

4.3. Responsibilities.

4.3.1. The 26 OWS will perform the OWS responsibilities stated in DAFMAN 15-129, as supplemented. Additionally, the 26 OWS will:

4.3.1.1. Perform a continuous Terminal METWATCH for Maxwell AFB.

4.3.1.2. Issue all watches, warnings and advisories for both Maxwell AFB and Gunter Annex, when Maxwell Weather is closed.

4.3.2. Maxwell Weather will perform duties IAW DAFMAN 15-129 and the contract. Additionally, Maxwell Weather will:

4.3.2.1. Issue all watches, warnings and advisories when the airfield is open and during severe weather when SWAP is activated.

4.3.2.2. Notify the 26 OWS after issuing, modifying, or canceling all observed warnings and advisories for Maxwell AFB.

4.3.2.3. Convey all weather watches, warnings, and advisories verbatim to base customers. No local modification and or supplementation is allowed on 26 OWS issued weather watches, warnings, and advisories.

4.4. Disseminating Weather Watches, Warnings, and Advisories.

4.4.1. The JET is the primary dissemination system for weather watches, warnings, and advisories on Maxwell AFB. All watches, warnings, and advisories will be disseminated via the JET. See **Attachment 4** for format and examples.

4.4.2. The 26 OWS/Maxwell WF will place a courtesy call, normally via the Integrated Weather Warning Capability (IWWC), to the Maxwell Command Post whenever they issue a watch or warning. The command post will disseminate the products to Maxell AFB organizations IAW **Attachment 6**.

4.4.3. Weather watches, warnings, and advisories will be identified numerically by month and sequential number of the product issued. For example, a weather warning number of "09-002" indicates the second warning (002) issued for the month of September (09). Sequential numbers for watches, warnings, and advisories increment independent of one another. The text of a weather watch, warning, or advisory will be worded so that all recipients may easily understand it.

4.5. Weather Watches. Weather watches are issued IAW DAFMAN 15-129.

Table 4.1. Forecast Weather Watch Criteria and Minimum Desired Lead-Times.

Watch Type	Criteria	Desired Lead Time
Tornadoes	Potential for Tornado or Funnel Cloud	60 Minutes prior to warning
Severe Thunderstorm	Potential for Damaging Winds ≥ 50 kts associated with thunderstorms and/or Damaging Hail ≥ 3/4 inch at Maxwell AFB	60 Minutes prior to warning
Moderate Thunderstorm	Potential for Strong Winds ≥ 35 kts and < 50 kts associated with thunderstorms and/or Large Hail ≥1/4 inch but < 3/4 inch at Maxwell AFB	60 Minutes prior to warning
Damaging Winds	Potential for surface winds not associated with thunderstorms ≥ 50 kts	60 Minutes prior to warning
Strong Winds	Potential for surface winds not associated with thunderstorms ≥ 35 kts but < 50 kts	60 Minutes prior to warning
Heavy Rain	Potential for Heavy Rain ≥ 2 inches within 6 hrs	60 Minutes prior to warning
Heavy Snow	Potential for Heavy Snow ≥ 2 inches within 12 hrs	60 Minutes prior to warning
Freezing Precipitation	Potential for Freezing Precipitation (Any Intensity)	60 Minutes prior to warning
Blizzard	Potential for Falling and or Blowing Snow lasting at least 3 hrs with winds ≥ 30kts and visibility ≤ 1/4mi	60 Minutes prior to warning
Lightning	Potential for Lightning within 10 NM*	30 Minutes
* Denotes deviation from DAFMAN 15-129		

4.6. Forecast Weather Warnings. Forecast warnings are issued IAW DAFMAN 15-129.

Table 4.2. Forecast Weather Warning Criteria and Minimum Desired Lead-Times.

Warning Type	Criteria	Desired Lead Time
Tornadoes	Tornado or Funnel Cloud	15 Minutes
Severe Thunderstorm	Damaging Winds ≥ 50 kts associated with thunderstorms and/or Damaging Hail ≥ 3/4 inch at Maxwell AFB	60 Minutes
Moderate Thunderstorm	Strong Winds ≥ 35 kts and < 50 kts associated with thunderstorms and/or Large Hail ≥1/4 inch but < 3/4 inch at Maxwell AFB	60 Minutes
Damaging Winds	Surface winds not associated with thunderstorms ≥ 50 kts	60 Minutes
Strong Winds	Surface winds not associated with thunderstorms ≥ 35 kts but < 50 kts	60 Minutes
Heavy Rain	Heavy Rain ≥ 2 inches within 6 hrs	60 Minutes
Heavy Snow	Heavy Snow ≥ 2 inches within 12 hrs	60 Minutes
Freezing Precipitation	Freezing Precipitation (Any Intensity)	60 Minutes
Blizzard	Falling and or Blowing Snow lasting at least 3 hrs with winds ≥ 30kts and visibility ≤ 1/4mi	60 Minutes

4.7. Forecast Weather Advisories. Forecast advisories are issued IAW DAFMAN 15-129.

Table 4.3. Forecast Weather Advisory Criteria and Minimum Desired Lead-Time.

Criteria	Desired Lead-Time
Temperature < 20° F	12 hours

4.8. Observed Weather Advisories. Maxwell Weather will issue an observed weather advisory for within 5 nm of the center of the Maxwell AFB runway complex when the criteria defined in **Table 4.4** occurs.

Table 4.4. Observed Weather Advisory Criteria and Minimum Desired Lead-Times.

Criteria	Desired Lead-Time
Cross Winds ≥ 25 Knots	First Observed
Cross Winds ≥ 15 Knots [1]	First Observed
Low Level Wind Shear Below 2000 feet [2]	First Observed
Note 1: When CAP-USAF is flying. Note 2: When reported by a pilot or observed on VAD wind profile	

4.9. Observed Weather Warnings. Maxwell Weather will issue an observed weather warning for within 5 nm of the center of the Maxwell AFB runway complex and/or within 5 nm of Gunter Annex when the criteria defined in **Table 4.5** occurs. When on duty, Maxwell Weather issues separate observed lightning warnings for within 5 nm of the center of Maxwell AFB and/or Gunter Annex. If Maxwell Weather is not on duty or evacuates the work center, 26 OWS will provide observed lightning warning support for Maxwell AFB. Since 26 OWS is limited to cloud-to-ground strike data from the National Lightning Detection Network, it does not have the capability to detect or observe other lightning strike occurrences (e.g., cloud-to-cloud lightning).

Table 4.5. Observed Weather Warning Criteria and Minimum Desired Lead-Times.

Criteria	Desired Lead-Time
Lightning within 5 NM of Maxwell AFB	First Observed
Lightning within 5 NM of Gunter Annex	First Observed
NOTE: Lightning warnings will be issued individually for within 5 NM of Maxwell AFB and within 5 NM of Gunter Annex.	

4.10. Lake Martin Resort.

4.10.1. Weather Watches. Weather watches are issued IAW DAFMAN 15-129. The 26OWS will issue a forecast weather watch for within 10 nm of Lake Martin Resort (Lat 32º 46' 30" N / Long 85º 49' 28" W) when the potential for the criterion defined in **Table 4.6** exists.

Table 4.6. Lake Martin Forecast Weather Watch Criteria and Minimum Desired Lead-Time.

Criterion	Desired Lead-Time
Lightning Within 10 NM of Lake Martin Resort [1]	30 minutes
Note 1: Deviates from standard DAFMAN 15-129 criteria	

4.10.2. Forecast Weather Warnings. Forecast weather warnings are issued IAW DAFMAN 15-129. 26 OWS will issue a forecast weather warning for within 5 nm of Lake Martin Resort when the criteria defined in **Table 4.7** occurs or is expected to occur.

Table 4.7. Lake Martin Forecast Weather Warning Criteria and Minimum Desired Lead-Times.

Warning Type	Criteria	Desired Lead Time
Tornadoes	Tornado or Funnel Cloud	15 Minutes
Severe Thunderstorm	Damaging Winds ≥ 50 kts associated with thunderstorms and/or Damaging Hail ≥ 3/4 inch at Lake Martin	60 Minutes
Moderate Thunderstorm	Strong Winds ≥ 35 kts and < 50 kts associated with thunderstorms and/or Large Hail ≥1/4 inch but < 3/4 inch at Lake Martin	60 Minutes
Damaging Winds	Surface winds not associated with thunderstorms ≥ 50 kts	60 Minutes
Strong Winds	Surface winds not associated with thunderstorms ≥ 35 kts but < 50 kts	60 Minutes

4.10.3. Observed Weather Warnings. 26 OWS will issue an observed weather warning for lightning within 5 nm of Lake Martin Resort when the criteria defined in **Table 4.8** occurs. Since 26 OWS is limited to cloud-to-ground strike data from the National Lightning Detection Network, it does not have the capability to detect or observe other lightning strike occurrences (e.g., cloud-to-cloud lightning).

Table 4.8. Lake Martin Observed Weather Warning Criteria and Minimum Desired Lead-Times.

Criterion	Desired Lead-Time
Lightning Within 5 NM of Lake Martin Resort	First Observed

4.10.4. Notification Hours. 26 OWS will issue all watches and warnings and notify Lake Martin 24 hours a day, 7 days a week.

4.11. Vigilant Warrior (Lake Jordan).

4.11.1. The Officer Training School (OTS) does periodic training at the Vigilant Warrior site (32º 38' 43" N 86º 16' 39" W). The 26 OWS has added the OTS Operation Control Center phone number (DSN (312) 493-9675) to their IWWC notification list for the communications site (MIDW), which is collocated with Vigilant Warrior at Lake Jordan.

4.11.2. Weather Watches. 26 OWS will issue a forecast weather watch for within 5 nm of Vigilant Warrior when the potential for the criteria defined in **Table 4.9** exists.

Table 4.9. Vigilant Warrior Forecast Weather Watch Criteria and Minimum Desired Lead-Times.

Watch Type	Criteria	Desired Lead Time
Tornadoes	Potential for Tornado or Funnel Cloud	60 Minutes prior to warning
Severe Thunderstorm	Potential for Damaging Winds ≥ 50 kts associated with thunderstorms and/or Damaging Hail ≥ 3/4 inch at Jordan Lake	60 Minutes prior to warning
Moderate Thunderstorm	Potential for Strong Winds ≥ 35 kts and < 50 kts associated with thunderstorms and/or Large Hail ≥1/4 inch but < 3/4 inch at Jordan Lake	60 Minutes prior to warning
Damaging Winds	Potential for surface winds not associated with thunderstorms ≥ 50 kts	60 Minutes prior to warning
Strong Winds	Potential for surface winds not associated with thunderstorms ≥ 35 kts but < 50 kts	60 Minutes prior to warning
Heavy Rain	Potential for Heavy Rain ≥ 2 inches within 12 hrs	60 Minutes prior to warning
Heavy Snow	Potential for Heavy Snow ≥ 2 inches within 12 hrs	60 Minutes prior to warning
Freezing Precipitation	Potential for Freezing Precipitation (Any Intensity)	60 Minutes prior to warning
Lightning	Potential for Lightning within 15 NM*	30 Minutes
An * denotes deviation from DAFMAN 15-129.		

4.11.3. Forecast Weather Warnings. Forecast warnings are issued IAW DAFMAN 15-129. 26 OWS will issue a forecast weather warning for within 5 nm of Vigilant Warrior when the criteria defined in **Table 4.10** occurs or is expected to occur.

Table 4.10. Vigilant Warrior Forecast Weather Warning Criteria and Minimum Desired Lead-Times.

Warning Type	Criteria	Desired Lead Time
Tornadoes	Tornado or Funnel Cloud	30 Minutes
Severe Thunderstorm	Damaging Winds ≥ 50 kts associated with thunderstorms and/or Damaging Hail ≥ 3/4 inch at Jordan Lake	120 Minutes
Moderate Thunderstorm	Strong Winds ≥ 35 kts and < 50 kts associated with thunderstorms and/or Large Hail ≥1/4 inch but < 3/4 inch at Jordan Lake	90 Minutes
Damaging Winds	Surface winds not associated with thunderstorms ≥ 50 kts	120 Minutes
Strong Winds	Surface winds not associated with thunderstorms ≥ 35 kts but < 50 kts	90 Minutes
Heavy Rain	Heavy Rain ≥ 2 inches within 12 hrs	90 Minutes
Heavy Snow	Heavy Snow ≥ 2 inches within 12 hrs	90 Minutes
Freezing Precipitation	Freezing Precipitation (Any Intensity)	60 Minutes
An * denotes deviation from DAFMAN 15-129.		

4.11.4. Observed Weather Warnings. Observed weather warnings are issued IAW DAFMAN 15-129. 26 OWS will issue an observed weather warning for within 5 nm of Vigilant Warrior when the criterion defined in **Table 4.11** occurs. Since 26 OWS is limited to cloud-to-ground strike data from the National Lightning Detection Network, it does not have the capability to detect or observe other lightning strike occurrences (e.g., cloud-to-cloud lightning).

Table 4.11. Vigilant Warrior Observed Weather Warning Criteria and Minimum Desired Lead-Times.

Criteria	Desired Lead-Time
Lightning within 15 NM of Vigilant Warrior	First Observed

4.12. Severe Weather Action Procedures (SWAP).

4.12.1. SWAP Responsibilities. These procedures are in place to ensure sufficient personnel are available during potential/actual severe weather events or during meteorological/operational events critical to mission success. Severe weather is defined as any weather phenomenon considered critical enough by the customer to require advance/special notice and subsequent actions to prevent serious injury or damage to personnel, property, or resources. It is imperative that timely and accurate weather watches, warnings and advisories are disseminated to all Maxell AFB agencies to ensure personnel and resource protection. There is a two-tier system with Maxwell Weather and 26 OWS sharing responsibilities for SWAP and resource protection.

4.12.2. Maxwell Weather Responsibilities. Maxwell Weather will perform the Exploitation Unit SWAP responsibilities as defined in DAFMAN 15-129, AFI 10-229, and AFMAN 10-206. More specifically, Maxwell Weather will accomplish the following actions:

4.12.2.1. Notification. The on duty weather forecaster will notify the SWAP standby member(s) according to the following guidance:

4.12.2.1.1. During weekday normal duty hours the duty forecaster will implement SWAP by notifying the Maxwell Weather manager and/or the SWAP standby member in the office or by phone whenever one or more conditions in Table 4.12 are met. It is likely that the above personnel are present in the weather office and do not need to be recalled/activated.

4.12.2.1.2. During weekend normal duty hours, the duty forecaster will implement SWAP by notifying the standby member and/or Weather Manager by phone whenever one or more conditions in **Table 4.12** are met.

4.12.2.2. Standby/Recall.

4.12.2.2.1. During Maxwell Weather non-duty hours, the 26 OWS will notify the Maxwell Command Post at DSN (312) 493-7474 of any Watch, Warning, or Advisory issued during METWATCH of the Maxell AFB operations area.

4.12.2.2.2. The weather manager will receive watches, warnings, and advisories by email from IWWC.

4.12.2.2.3. The weather manager and/or the SWAP standby member will analyze current atmospheric conditions and discuss the forecast with the 26 OWS Forecaster/Regional Weather Supervisor.

4.12.2.2.4. The SWAP will be activated whenever one or more of the warnings in **Table 4.12** are issued.

Table 4.12. Warnings Requiring Notification/Activation of SWAP Standby Member.

Tornado/Funnel Cloud[1]
Severe Thunderstorm[1]
Damaging Winds[1]
Freezing Precipitation
Snow Accumulation
Blizzard Conditions
Note 1: Watch or Warning. Note 2: In the event of unforeseen circumstances, such as a communications line failure, a critical equipment outage at either the 26 OWS or Maxwell Weather, Maxwell Weather will implement the SWAP at the 26 OWS' request. The 26 OWS, as the agency ultimately responsible for forecast watch/warning support, will have this prerogative in the interest of Maxell AFBresource protection and flight safety.

4.12.3. Activation. The weather forecaster on duty will discuss the meteorological situation, manning requirements, and the recall of additional personnel (or place on standby) with the SWAP standby member. (If the SWAP standby member is unavailable, coordinate with the Maxwell Weather manager). If deemed necessary, the SWAP standby member will report to

the weather office no later than 30 minutes after notification by the duty forecaster or as soon as possible. Once the SWAP standby member has arrived, they will assist in evaluating the situation, determine the need to recall additional personnel, and execute the SWAP duties/responsibilities in **Table 4.13**.

Table 4.13. SWAP Duties/Responsibilities.

Duty Forecaster	
1.	Notify/recall SWAP Standby Member (Team Chief). Phone numbers can be found on the Recall Roster.
2.	Constantly coordinate with the 26 OWS on Watches/Warnings.
3.	Eyes Forward - Intensify monitoring of local and area weather conditions to enhance METWATCH and MISSIONWATCH.
4.	Notify appropriate agencies of the issuance of Watches/Warnings. Note: Call Command Post and notify them immediately when a tornado is occurring or is within 10 minutes of occurring within 5NM of the Maxwell and/or Gunter and they will activate the siren(s).
5.	Advise senior base leadership of the situation when warranted.
6.	Review SWAP Standby Member/Team Chief checklist and begin any duties, as necessary, until the member arrives.
7.	Conduct a concise forecast discussion of the current situation to apprise SWAP Standby Member/Team Chief upon their arrival.
8.	If necessary, augment weather observations in accordance with METAR and SPECI criteria and AFMAN 15-111.
9.	Issue observed Warnings/Advisories.
10.	Review PIREPs, SIGMETs, and area NWS products for severe weather reports. If applicable, notify the 26 OWS and incorporate data into products.
11.	Update Mission Weather Products (MWPs) as needed.
12.	Work closely with the SWAP members.
13.	Provide inputs to post-event OPREP-3 report (if required).
SWAP Standby Member/Team Chief	
1.	Report to the Maxwell Weather office within 30 minutes of initial notification. Determine if the situation warrants the recall/stand-by placement of additional personnel.
2.	Upon arrival, receive initial forecast discussion from Duty Forecaster.
3.	When time allows, conduct a METCON with the 26 OWS Forecaster/Zone Supervisor.
4.	Assume radar operator duties if other personnel are unavailable.
5.	Ensure duty positions are delegated and members are performing assigned tasks (combine duty positions when necessary). a. Forecaster b. Radar operator

6.	Ensure the following tasks are accomplished on a recurring basis: a. Recall additional personnel if needed. b. Adjust duties as deemed necessary. c. As requested, keep senior base leadership, command post and customers apprised of latest developments. d. Keep personnel focused on assigned tasks. e. Ensure all applicable watches/warnings/advisories are issued and notification has been accomplished. f. Review all forecast products for accuracy and horizontal consistency (e.g., watches/warnings/advisories, TAFs, and MWPs). g. Provide meteorological expertise and guide decision-making process.
7.	Conduct post-event review and discussion to provide team members with feedback (positive and negative).
8.	Consolidate inputs and coordinate with the 26 OWS for OPREP-3 or HOMELINE report and provide to Maxwell Command Post (if required).
RADAR Operator	
1.	Interrogate storms and related phenomena using RADAR and, during thunderstorms, lightning detection data.
2.	Keep Duty Forecaster and Team Chief informed of local severe activity.
3.	During tornado and thunderstorm events, provide the Duty Forecaster with storm positions and movements.
4.	During thunderstorm events, advise the SWAP members when thunderstorms are within 10nm and 5nm.
5.	Answer phones and questions. Prioritize calls for the Duty Forecaster and Team Chief. Direct calls from unofficial sources to Public Affairs.
6.	Answer PMSV calls.
7.	Assist other team members as needed.
8.	Archive data if deemed necessary.
9.	Work closely with other SWAP members. Allow them to accomplish tasks, which will free the duty forecaster to handle critical tasks such as watch/warning/advisory issuance/notification, MWP amendments and coordination with the 26 OWS.
10.	If severe criteria occurs, archive radar data.
11.	Provide inputs to DAF Report (if required).

4.12.4. Post Event Procedures. If severe weather actually occurs, the following procedures will be executed, if necessary, by Maxwell Weather:

4.12.4.1. DAF Reporting. When significant weather occurs and results in Class A and/or B damage, base closure, or mission degradation., the following information will be provided to the Maxwell Command Post: actual severe weather experienced; Terminal Aerodrome Forecast (TAF) valid at the time of occurrence; any watches, warnings and or advisories issued to include actual and desired lead-time; and operational status of meteorological equipment.

4.12.4.2. MAJCOM Commander Critical Information Requirements (CCIR) Reporting. When weather related events meet Class C, D, and E damage criteria occur and result in

aircraft damage, injuries, significant impact to student training, cancellation of flying or technical training, and restricted duty hours or delays in reporting for extended time, Maxwell Weather will provide any requested weather information to the command post for inclusion in a CCIR report at the commander's discretion.

4.12.4.3. Maxwell Weather will notify the 26 OWS and AETC/A3OW of the DAF Report or HOMELINE reports and information submitted for inclusion.

4.12.4.4. Maxwell Weather will request that the 26 OWS provide required information, if needed, by performing a weather data save.

4.12.4.5. Maxwell Weather will provide the 26 OWS with severe weather reports not normally available through standard observations. Reports will be passed immediately after fulfilling any local distribution requirement. If this is not possible, Maxwell Weather will pass the reports as soon as possible so the 26 OWS can use the reports during post-analysis and verification.

4.12.4.6. If the event is to be used as one of the semi-annual SWAP tests, Maxwell Weather will complete a memorandum for record documenting the event. Also contact 26 OWS flight leadership and provide them with the memorandum or an e-mail for their records.

4.12.4.7. Maxwell Weather personnel's home phone numbers and cell phone numbers will be reviewed monthly or when a change in personnel occurs to ensure the most current numbers are available for SWAP.

4.12.5. 26 OWS Responsibilities. The 26 OWS routinely schedules adequate manning resources for each duty shift to respond to severe weather situations within the AOR. The OWS Operations Floor SDOs/SDNCOs continuously assess severe weather threats and shift manning, as needed to address additional workload requirements for locations.

SHAMEKIA N. TOLIVER, Colonel, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFPD 15-1, *Weather Operations*, 28 May 2024

AFMAN 10-206, *Operational Reporting (OPREP)*, 18 June 2018, incoporting Change 1, dated 1 September 2020

AFMAN 13-204V1, *Management of Airfield Operations*, 22 July 2020

AFMAN 15-111, *Surface Weather Observations*, 12 March 2019

AFMAN 15-124, *Meteorological Codes*, 16 January 2019

DAFMAN 15-129, *Air and Space Weather Operations*, 7 September 2023

DAFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards,* 25 March 2022

DAFI 10-2501 MAXWELLAFBSUP 24 June 2024

FAA Order 7110.65, *Air Traffic Control*, 20 April 2023

Prescribed Forms

Maxwell AFB Form 27, *Aircrew Briefing Log*

Maxwell AFB Form 28, *Pilot-to-Metro-Service Log*

Adopted Forms

DD Form 175-1, *Flight Weather Briefing*

DAF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

ACFT MISHAP—Aircraft Mishap

AETC—Air Education and Training Command

AFW—Air Force Weather

AFWA—Air Force Weather Agency

ALSTG—Altimeter

AMD—Amendment

AOL—Alternate Operating Location

ATC—Air Traffic Control

AWN—Automated Weather Network

BKN—Broken

BWW—Basic Weather Watch

CAP – USAF—Civil Air Patrol – United States Air Force

CB—Cumulonimbus

CBRNE—Chemical, Biological, Radiological, Nuclear, and High-Yield Explosive

CCIR—Commander Critical Information Requirements

COOP—Continuity of Operations Plan

CU—Characterization Unit

CWW—Cooperative Weather Watch

DLT—Desired Lead Time

EM—Emergency Management

EOC—Emergency Operations Center

EU—Exploitation Unit

FLIP—Flight Information Publication

GPS—Global Positioning System

HURCON—Hurricane Condition

ICAO—International Civil Aviation Organization

IRC—Instrument Refresher Course

IWWC—Integrated Weather Warning Capability

JET—Joint Environmental Toolkit

KT—Knot

LLWS—Low-Level Wind Shear

LNS—Lightning Notification System

MEF—Mission Execution Forecast for Flight Missions

MEFP—Mission Execution Forecast Process

METAR—Meteorological Aviation Report

METCON—Meteorological Conference or Discussion

METWATCH—Meteorological Watch

METSAT—Meteorological Satellite

MISSIONWATCH—Mission Meteorological Watch

MWP—Mission Weather Product

NHC—National Hurricane Center

NIMA—National Imagery Mapping Agency

NLT—No Later Than or Negative Lead Time

NM—Nautical Mile

NWS—National Weather Service

OVC—Overcast

OWS—Operational Weather Squadron

PIREP—Pilot Weather Report

PMSV—Pilot –to-Metro Service

QRC—Quick Reaction Checklist

RVR—Runway Visual Range

SDNCO—Senior Duty Noncommissioned Officer

SDO—Senior Duty Officer

SOP—Standing Operating Procedures

SPECI—Special Observation

SWAP—Severe Weather Action Procedures

TAF—Terminal Aerodrome Forecast

TAS—Telephone Alert System

TEMPO—Temporary

TC-TAP—Tropical Cyclone Threat Assessment Product

VFR—Visual Flight Rules

VT—Valid Time

WA—Weather Advisory

WSD—Weather Support Document

WSR-88D—NEXRAD (Next Generation Doppler Radar)

WW—Weather Warning

WWA—Watches, Warnings, and Advisories

Office Symbols

26 OWS—26th Operational Weather Squadron

42 ABW—42d Air Base Wing

42 OMRS/SGXB—Bioenvironmental Engineering

42 OSS—42d Operations Support Squadron

703d HS—703d Helicoptert Squadron

908 AW—908th Flying Training Wing

OTS/OCC—Officer Training School/Officer Candidate Course

Terms

26th Operational Weather Squadron (26 OWS) – A facility located at Barksdale AFB LA that provides primary Terminal Airdrome Forecasts, Weather Watches and forecasted Warnings and Advisories. The 26 OWS is the central forecasting facility for South-eastern CONUS weather operations.

Aircraft Mishap – Term used to denote any event resulting in damage to, or destruction of any aircraft to include lightning strikes, inadvertent departure from the paved runway or taxiway surface, aircraft or Aerospace Ground Equipment (AGE) fires, and forced landings due to in-flight emergencies.

Amendment (AMD) – Used as a message modifier when transmitting an aerodrome forecast amendment.

Augmentation – Augmentation is the process of manually adding or editing data in an observation generated by the FMQ-19 by either supplementing (adding) or back-up (editing).

Basic Weather Watch (BWW) – A BWW is normally conducted by Maxwell Weather when the FMQ-19 is in augmentented or manual mode because they cannot monitor the atmospheric conditions continuously due to other weather duties. During a BWW, weather forecasters will recheck weather conditions, at intervals not to exceed 20 minutes since the last observation/recheck, to determine the need for a SPECI observation.

Climatology – The historical records of weather conditions measured or observed at a specific location. Some data go back over 100 years, but generally, a 10- to 25-history is more common. Climatology is useful in planning operations beyond 5 to 7 days. It usually describes the average (or mean) conditions such as average high and low temperatures and extremes.

Cooperative Weather Watch – Air Traffic Control personnel notify the forecaster of significant changes from reported conditions, as duty priorities permit.

Desired Lead-time (DLT) – The amount of advance notice a supported agency desires before the onset of a particular weather phenomenon.

Eyes Forward – Maxwell Weather forecasters are the eyes forward for the forecasters in the Operational Weather Squadron (OWS) and integrate weather radar data, meteorological satellite imagery, lightning detection readouts, and non-standard weather data systems (vertical profilers, mesonet data, etc.) to create an integrated weather picture and near-term trend forecasts for the OWS.

Forecast Weather Advisory (FWA) – A weather advisory issued when the customer requires advance notification of an impending weather condition with sufficient time to allow for protective actions.

Integrated Weather Warning Capability (IWWC) – Automated system used by the 26 OWS to automatically disseminate weather watches, warnings, and advisories over the JET, by phone, and over their web page.

Issue Time – The time when an agency is notified of a watch, warning, or advisory. When more than one agency is notified, the issue time is the time the last agency is notified. Follow-up notifications are not considered when determining issue time.

Joint Environmental Toolkit (JET) – JET is a web based weather dissemination system. Products available for users on JET include weather warnings, weather watches, weather advisories, live sensor data, surface weather observations, and terminal aerodrome forecasts. All data is site-specific for Maxwell AFB and Gunter Annex. Use of JET requires a username and password which can be provided by Maxwell Weather.

Limited Weather Observation – The limited weather observation is a surface weather observation taken by certified non-weather personnel. Units that take limited weather observations must relay these to Maxwell Weather in support of the Cooperative Weather Watch program.

Maxwell Weather – The weather facility and or weather personnel located in Bldg 844, Maxwell AFB AL. Maxwell Weather provides direct operational support at the tactical/base level.

METAR Observation – Meteorological Aviation Report. A routine, hourly surface weather observation. It contains a report of wind, visibility, runway visual range, present weather, sky condition, temperature, dew point, and altimeter setting. In addition, significant remarks are appended to the METAR observation.

METWATCH – Monitoring aerospace weather for a route, area, or terminal and advising concerned organizations when phenomena that could affect their operations or pose a hazard to life or property are occurring or about to occur.

Mission Weather Product (MWP) – Any weather product or group of weather products generated by Maxwell Weather that is integrated into the military decision making process. MWPs may be planning or execution products and are not limited to aviation missions.

MISSIONWATCH – The monitoring of aerospace weather for a specific mission (i.e., ground, air, or space) and informing supported agencies when unforecast mission-limiting phenomena could impact operations.

Observed Weather Advisory (OWA) – A weather advisory issued when a particular weather event first occurs and the customer does not require advanced notification of the observed weather phenomena.

Operational Weather Squadron (OWS) – An organization comprised of management, technician, and training personnel responsible for providing regional weather support. Their mission is to produce theater-scale tailored weather forecast products and services to customers within their area of responsibility.

Pilot Report (PIREP) – A report of in-flight weather provided by an aircrew member.

Pilot to Metro Service (PMSV) – A radio system with dedicated frequency (342.3 MHz) for contact between airborne pilots and weather facilities.

Severe Weather – Any weather condition that poses a hazard to property or life.

SPECI Observation – An unscheduled observation taken when significant changes in weather elements meet special criteria. All SPECIs shall be made as soon as possible after the relevant criteria are observed.

Surface Weather Observation – An evaluation of one or more meteorological elements that describe the state of the atmosphere at the location where the observation is taken. Observations are taken in accordance with AFMAN 15-111, *Surface Weather Observations*, and disseminated locally and via longline communications circuits during hours of operation.

Temporary (TEMPO) – Indicates temporary fluctuations in forecast conditions. Conditions described will last for less than one hour and cover less than half of the designated TEMPO period.

Terminal Aerodrome Forecast (TAF) – A weather forecast prepared by the 26th Operational Weather Squadron (OWS) composed of required weather elements for Maxwell AFB airfield and covers a 24-hour period. Forecast elements in the body of the forecast text refer to the area within 5 NM of the center of the aerodrome complex. Operationally significant elements outside this area are included in remarks (e.g., TS OMTNS or VCTS). The term VC (vicinity) refers to the area between 5 NM and 10 NM of the aerodrome complex.

Weather Advisory (WA) – A special notice provided to a supported agency when an established weather condition that could affect its operation is occurring or is expected to occur.

Weather Surveillance Radar-1988, Doppler (WSR-88D) – A Doppler weather radar located at Carrville AL designed to: provide range and azimuth information on precipitation within a radius of 248 nautical miles (NM), wind velocity, storm structure, hail detection, and total precipitation within 124 NM, and tornado detection within 60 NM. The East Alabama WSR-88D is part of the national Doppler weather radar network, and is controlled by the National Weather Service office in Birmingham AL. Commonly referred to as NEXRAD.

Weather Warning (WW) – A special notice to notify operational commanders when an established weather condition of such intensity as to pose a hazard to life or property is occurring or is expected to occur. Weather warnings provide concise information outlining environmental threats and are used by operational commanders to make force protection decisions.

Weather Watch – A special notice provided to supported customers that alerts them of a potential for weather conditions of such intensity as to pose a hazard to life or property for which the customer must take protective action.

Attachment 2

SPECIAL/LOCAL OBSERVATION CRITERIA

A2.1. Special (SPECI) Criteria.

A2.1.1. Visibility. Surface visibility as reported in the body of the report decreases to less than or, if below, increases to equal or exceed.

A2.1.1.1. 3 miles (AFMAN 15-111).

A2.1.1.2. 2 miles (AFMAN 15-111 and DoD FLIPS).

A2.1.1.3. 1 5/8 miles (DoD FLIPS).

A2.1.1.4. 1 1/2 miles (DoD FLIPS).

A2.1.1.5. 1 1/4 miles (DoD FLIPS).

A2.1.1.6. 1 1/8 miles (DoD FLIPS) (When Approach Lighting Systems not operational.).

A2.1.1.7. 1 mile (AFMAN 15-111 and DoD FLIPS).

A2.1.1.8. 3/4 mile (DoD FLIPS).

A2.1.1.9. 5/8 mile (DoD FLIPS).

A2.1.1.10. 1/2 mile (DoD FLIPS).

A2.1.1.11. 1/4 mile (AFMAN 15-111)

A2.1.2. Ceiling. The ceiling (rounded off to reportable values) forms or dissipates below, decreases to less than, or if below, increases to equal or exceed.

A2.1.2.1. 3,000 feet (AFMAN 15-111).

A2.1.2.2. 2,000 feet (AFMAN 15-111).

A2.1.2.3. 1,500 feet (AFMAN 15-111).

A2.1.2.4. 1,000 feet (AFMAN 15-111).

A2.1.2.5. 800 feet (AFMAN 15-111).

A2.1.2.6. 700 feet (AFMAN 15-111).

A2.1.2.7. 600 feet (DoD FLIPS).

A2.1.2.8. 500 feet (AFMAN 15-111).

A2.1.2.9. 400 feet (DoD FLIPS).

A2.1.2.10. 300 feet (AFMAN 15-111).

A2.1.2.11. 200 feet (DoD FLIPS).

A2.1.2.12. 100 feet (AFMAN 15-111).

A2.1.3. Sky Condition. A layer of clouds or obscuring phenomena aloft is observed below 600 feet and no layer aloft was reported below this height in the previous METAR or SPECI.

A2.1.4. Wind Shift. Wind direction changes by 45 degrees or more in less than 15 minutes and the wind speed is 10 knots or more throughout the wind shift.

A2.1.5. Squall. When squalls occur.

A2.1.6. Volcanic Eruption. When eruption first noted.

A2.1.7. Thunderstorm (occurring at the station). Begins or ends.

A2.1.8. Precipitation. **NOTE:** Except for freezing rain, freezing drizzle, ice pellets, and hail, a SPECI is not required for changes in type (e.g., drizzle changing to snow grains) or the beginning or ending of one type while another is in progress (e.g., snow changing to rain and snow).

A2.1.8.1. Hail begins or ends.

A2.1.8.2. Freezing precipitation begins, ends, or changes intensity.

A2.1.8.3. Ice pellets begin, end, or change in intensity.

A2.1.8.4. Any other type of precipitation begins or ends.

A2.1.9. Tornado or Funnel Cloud.

A2.1.9.1. Is observed.

A2.1.9.2. Disappears from sight or ends.

A2.1.10. Runway Visual Range (RVR).

A2.1.10.1. Prevailing visibility is first observed less than or equal to 1SM, and again when prevailing visibility goes above 1SM.

A2.1.10.2. RVR for active runway decreases to less than, or if below, increases to equal or exceed

A2.1.10.2.1. 6000 feet (AFMAN 15-111)

A2.1.10.2.2. *5500 feet (DoD FLIPS) * - When Approach Lighting System not operational.

A2.1.10.2.3. 5000 feet (AFMAN 15-111)

A2.1.10.2.4. 4000 feet (DoD FLIPS)

A2.1.10.2.5. 3500 feet (DoD FLIPS).

A2.1.10.2.6. 2400 feet (AFMAN 15-111 and DoD FLIPS)

A2.1.10.2.7. 2000 feet (AFMAN 15-111)

A2.1.10.3. RVR is first determined as unavailable (RVRNO) for the runway in use, and when it is first determined that the RVRNO report is no longer applicable, provided conditions for reporting RVR exist.

A2.1.11. Upon Resumption Of Observing Function. A SPECI observation will be taken as soon as possible after the weather forecaster returns to duty following a break in observing coverage or augmentation unless a record observation is filed during that 15-minute period.

A2.1.12. Aircraft Mishap. Disseminated regardless of operating mode, i.e. augment, auto, or manual.

A2.1.13. Miscellaneous. Any other meteorological situation that in the weather forecaster's opinion is critical.

A2.2. LOCAL Criteria. Single element LOCALs will only be taken for altimeter setting changes during back-up of the FMQ-19 pressure sensor. LOCAL altimeter setting observations will be taken at an interval not to exceed 35 minutes when there has been a change of 0.01 inch Hg (0.3 hPa) or more since the last ALSTG value.

Attachment 3

TAF SPECIFICATION AND AMENDMENT CRITERIA

A3.1. As a minimum, TAFs will specify time of occurrence to the nearest hour (and minute as appropriate), the duration, and intensity of the standard criteria listed in **Table A3.1**. TAFs will be amended when the criteria occur or are expected to occur but were not specified in the forecast. Also, the TAF will be amended when the criteria is specified in the forecast but are no longer occurring or expected to occur.

Table A3.1. TAF Specification/Amendment Criteria.

Forecast Element/Occurrence	TAF Amendment Criteria	
CEILING or VISIBILITY observed or later expected to decrease to less than, or if below, increase to equal or exceed:	Category	Limits (Ceiling/Visibility)
	E	≥ 2000 ft/3 miles
	D	< 2000/3 and ≥ 1000/2[1]
	C	< 1000/3 and ≥500/2[1]
	B	< 500/2 and ≥200/½
	A	< 200/½
SURFACE WINDS	SPEED The difference between the predominant wind speed and the forecast wind speed is ≥ 10 knots and/or the difference between the observed gusts is ≥ 10 knots from the forecast gust.	
	DIRECTION A change > 30 degrees when the predominant wind speed or gusts are expected to be over 15 knots.	
ICING not associated with thunderstorms, from the surface to 10,000 feet AGL.	The beginning or ending of icing first meets, exceeds, or decreases below moderate or greater thresholds and was not specified in the forecast.	
TURBULENCE (for Cat II aircraft), not associated with thunderstorms, from the surface to 10,000 feet AGL.	The beginning or ending of turbulence first meets, exceeds, or decreases below moderate or greater thresholds and was not specified in the forecast.	
WARNINGS and ADVISORIES (TAF-amendable advisories including Low-Level Wind Shear) Watches may or may not be specified in a TAF depending on situation.	Occur, or are expected to occur, during the forecast period, but were not specified in the forecast. Were specified in the forecast, but are no longer occurring or expected to occur during the forecast period.	
Altimeter Setting	Altimeter setting meets or exceeds 31.00 INS and was not specified in the forecast Altimeter setting, if above, drops below 31.00 INS and was not specified during the forecast period Altimeter setting drops below 28.00 INS and was not specified in the forecast	

	Altimeter setting, if below 28.00 INS, increases above 28.00 INS and was not specified in the forecast"
Forecast Weather Advisory Criteria issued for amendable TAF criteria.	Occur, or are expected to occur during the forecast period, but were not specified in the forecast Specified in the forecast but are no longer expected to occur during the forecast period
THUNDERSTORMS	Incorrect forecast start or end times
TEMPO conditions	TEMPO conditions become predominant conditions. TEMPO conditions do not occur as forecast. TEMPO conditions are no longer expected to occur.
PREDOMINANT conditions	Forecast conditions occur before the beginning of the specified period of change and are expected to persist. Forecast conditions do not occur within 30 minutes after the specified time. Forecast conditions are no longer expected to occur.
REPRESENTATIVE conditions	Forecast conditions are considered unrepresentative of existing or forecast conditions and amending the forecast improves safety, flight planning, operations efficiency, or assistance to in-flight aircraft.
Note 1: Forecast ceiling/visibility category is determined by the lower ceiling or visibility value.	

Attachment 4

JET DISSEMINATION FORMATS

A4.1. Surface Weather Observation. This is a partial list of the most common elements in a surface weather observation. For more information on the surface weather observation code (METAR), see AFMAN 15-111, *Surface Weather Observations*.

Figure A4.1. Surface Weather Observation Code Breakdown.

METAR KMXF 121855Z AUTO 12003KT 5SM -TSRA SCT010CB BKN040 1 2 3 4 5 6 7 8 21/07 A2987 RMK AO2 TS 5NE MOV NE 9 10 11 12 13 14	
1 - Type of observation (METAR or SPECI) 2 - Station identifier for Maxwell 3 - Valid date/time 4 - Automated observation Indicator 5 - Wind direction and speed (knots) 6 - Visibility (miles) 7 - Weather occurring at time of observation	8 - Sky condition (clouds) 9 - Temperature in Celsius 10 - Dew Point in Celsius 11 - Altimeter in inches 12 - Indicates remarks will follow 13 - Augmented unit indicator 14 - Thunderstorm remark

A4.2. Terminal Aerodrome Forecast (TAF). This is a partial list of the most common elements in a terminal aerodrome forecast. For more information on the terminal aerodrome forecast code, see AFMAN 15-124, *Meteorological Codes*.

Figure A4.2. Terminal Aerodrome Forecast Code Breakdown.

```
TAF KMXF 1220/1402 24006KT 8000 -SHRA BKN030 OVC200 QNH3022INS
 1   2      3        4     5    6        7            8
    TEMPO 1220/1222 VRB15G25KT 4800 -TSRA BKN010CB OVC020
      9      10
    BECMG 1311/1312 11004KT 9000 -SHRA FEW030 BKN080 OVC200
      11     12
    QNH3019INS T18/1320Z T12/1309Z
                  13        14
```

1 - TAF indicates terminal aerodrome forecast
2 - KMXF is station identifier for Maxwell
3 - Valid period date/time
4 - Wind direction and speed (knots)
5 - Visibility (meters)
6 - Current Weather Conditions
7 - Sky Condition (clouds)
8 - Altimeter in Inches
9 - TEMPO indicates conditions will vary for the valid time period
10 - Valid time period for TEMPO conditions
11 - BECMG indicates conditions will change during time indicated
12 - Time period BECMG group will occur
13 - Maximum temperature during forecast valid period and time of occurrence
14 - Minimum temperature during forecast valid period and time of occurrence

A4.3. Pilot Report (PIREP). This is a partial list of the most common elements in a pilot report. For more information on the pilot report code, see AFMAN 15-124, *Meteorological Codes*.

Figure A4.3. Pilot Report (PIREP) Code Breakdown.

```
KMXF UA /OV KMXF270020/TM 1320/FL230/TP C130/SK BKN040-
            1          2      3      4        5
    TOP080/WX FV02SM -RA BR/TA M22/WV 27040KT/TB LGT CHOP 200-
              6             7        8              9
    240/IC NEG/RM LLWS -10KT SFC-010 DURGC RWY 33 KMXF;
       10   11                  12
```

1 - Location of aircraft
2 - Time of report
3 - Flight level
4 - Aircraft type
5 - Sky Condition (clouds)
6 - Visibility (miles) and weather occurring
7 - Temperature in Celsius at flight level
8 - Wind direction and speed in knots at flight level
9 - Turbulence
10 - Icing
11 - RM indicates remarks will follow
12 - Remarks

A4.4. Weather Watches, Warnings, and Advisories.

Figure A4.4. Weather Watch, Warning, and Advisory Code Breakdown.

Weather Watch 01-005 for Maxwell AFB (KMXF) Valid 10/0400Z
1 2 3
(9/2200L) to 10/1200Z (10/0600L) Potential for Freezing
3 4
Precipitation (Any Intensity) is in effect at Maxwell AFB-
4
Gunter Annex.
4
Issued at 1/9/2011 1535Z
5

1 - Product type - Weather watch, warning or advisory 2 - Watch, warning, or advisory number 3 - Valid period. UFN indicates valid until further notice	4 - Text of watch, warning, or advisory 5 - Date/time product was issued

Figure A4.5. Example Weather Watches, Warnings, and Advisories.

WEATHER WATCH

Weather Watch 02-004 for Maxwell AFB (KMXF) Valid 6/2200Z (6/1600L) to 7/0300Z (6/2100L) Potential for Tornado exists within 5 nm. of Maxwell AFB-Gunter Annex.
Issued 2/6/2011 1508Z

WEATHER WARNING

Weather Warning 04-018 for Maxwell AFB (KMXF) Valid 13/1700Z (13/1200L) to 13/2100Z (13/1600L) Large Hail >= 3/4 in. maximum expected 1 in. is forecasted to occur at Maxwell AFB-Gunter Annex. Damaging Winds greater than or equal to 50 kts. maximum expected 60 kts. are forecasted to occur at Maxwell AFB-Gunter Annex.
Issued 4/13/2011 1412Z

WEATHER ADVISORY

Weather Advisory Valid 23/1349Z (23/08491) UFN Observed Low Level Wind Shear occurring below 2000 ft AGL. at Maxwell AFB
Issued 4/23/2011 1349Z

Attachment 5

WEATHER IMPACTS ON SUPPORTED UNITS

A5.1. The following table lists specific weather criteria, customers supported by Maxwell Weather, the weather impacts on their operations, and the actions customers take to mitigate or protect operations from those impacts. Weather watches alert customers to the potential for severe weather and allow them to prepare to take the actions noted under each warning criteria.

Table A5.1 Weather Impacts to Supported/Base Customers.

TORNADO WARNING		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURE**
All	Danger to life and property.	Take shelter immediately.
DAMAGING WINDS ≥ 50 KNOTS WARNING		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
CAP-USAF	Cancels flight ops.	Reschedule flight ops. Hangar or tie down aircraft. Head into wind if uninstalled engines or cannot be tied down. Ensure duct plugs and shields are installed, raise flaps to full up. Fuel to fullest capacity, close all doors and hatches.
42 OSS/AOT	1. Aircraft/equipment may be damaged. 2. Loose objects may cause damage.	1. Secure open canopies 2. Hangar as many aircraft as possible 3. Tie down aircraft 4. Secure loose items
AU	Cancels outdoor training	Direct students inside hardened buildings
OSS (Tower)	Threat to personnel.	Evacuate control tower.
42 SFS	Threat to personnel training/Responding patrols	Cancel firing for Combat Arms. If visibility is significantly decreased, limit patrol movement to response only Close CVI/Installation gates at discretion of SF Operations Superintendant or Operations Officer. Threat to Equipment" "Skywatch Towers are brought down.
Vigilant Warrior	Threat to personnel.	Vacate open areas and take cover in hardened buildings
Lake Martin Rec Area	Threat to recreation area and campers	Vacate open areas and Lake Martin and take shelter in hardened buildings
HAIL ≥ 3/4 INCH WARNING		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
CAP-USAF	Cancels flight ops.	Reschedule flight ops. Hangar as many aircraft as possible
42 OSS/AOT	Potential damage to aircraft	1. Secure open canopies 2. Hangar as many aircraft as possible

AU	Cancels outdoor training	Direct students inside hardened buildings
OSS (Tower)	Threat to personnel.	Evacuate control tower at discretion of Watch Supervisor.
42 SFS	Threat to personnel training	Direct students inside hardened facilities
42SFS	Threat to personnel Training/Military Working Dogs (MWDs)	1. Secure MWDs kennel runs exterior dog doors. 2. Limit utilization for emergency responses only via coordination with the kennel master/trainer and at the discretion of operations superintendent/officer. 3. Cancel firing for Combat Arms
42 SFS	Threat to response vehicles	Possibly parked under hardened area parking structure until dispatched for emergency
42 SFS	Threat to personnel	Possibly shut down CVI/gates at discretion of operations superintendent/officer
42 SFS	Threat to equipment	Skywatch towers are brought down.
Vigilant Warrior	Threat to personnel.	Vacate open areas and take cover in hardened buildings
Lake Martin Rec Area	Threat to recreation area and campers	Vacate open areas and Lake Martin and take shelter in hardened buildings
HIGH WIND ≥ 35 KNOTS BUT < 50 KNOTS WARNING		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
CAP-USAF	Cancels flight ops.	Reschedule flight ops. Hangar aircraft
42 OSS/AOT	1. Aircraft/equipment may be damaged. 2. Loose objects may cause damage.	1. Tie down aircraft or move into hangars. 2. Move loose equipment to storage. 3. Secure loose objects and check area for security. 4. Secure open canopies
AU	Reschedules outdoor training	Students train inside hardened buildings
Base Civil Engineer	Increased unscheduled maintenance, damage to trees, buildings. Direction Change: Runway Change	1. Reschedule routine work. 2. Remove fallen trees, repair buildings
42ABW/SFS	Loose objects may cause damage.	Advise personnel to secure all objects. Notify housing areas.
42 SFS	Threat to personnel training	Direct students inside hardened facilities Cancel Firing for Combat Arms
42SFS	Threat to Military Working Dogs (MWDs)	1. Secure MWDs kennel runs exterior dog doors. 2. Limit utilization for emergency responses only via coordination with the

		kennel master/trainer and at the discretion of operations superintendent/officer.
42 SFS	Threat to response vehicles	Possibly park vehicles under hardened area parking structure until dispatched for emergency
42 SFS	Threat to personnel	Possibly shut down CVI/gates at discretion of operations superintendent/officer
42 SFS	Threat to equipment	Skywatch towers are brought down.
Vigilant Warrior	Threat to personnel.	Vacate open areas and take cover in hardened buildings
Lake Martin Rec Area	Threat to recreation area and campers	Vacate open areas and Lake Martin and take shelter in hardened buildings
HAIL ≥ 1/2 BUT < 3/4 INCH WARNING		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
CAP-USAF	Cancels flight ops.	Reschedule flight ops. Hangar aircraft
42 OSS/AOT	Aircraft/equipment may be damaged.	1. Move as many aircraft as possible into hangars. 3. Move loose equipment to storage. 4. Secure loose objects and check area for security. 5. Secure open canopies
AU	Reschedules outdoor training	Students train inside hardened buildings
Base Civil Engineer	Increased unscheduled maintenance, damage to trees, buildings. Direction Change: Runway Change	1. Reschedule routine work. 2. Remove fallen trees, repair buildings
42ABW	Outside objects may be damaged.	Advise personnel to secure all objects. Notify housing areas.
42 SFS	Threat to personnel training	Direct students inside hardened facilities
42SFS	Threat to Military Working Dogs (MWDs)	1. Secure MWDs in kennel runs and close MWD exterior dog doors. 2. Limit utilization for emergency responses only via coordination with the kennel master/trainer and at the discretion of operations superintendent/officer.
42 SFS	Threat to response vehicles	Possibly parked under hardened area parking structure until dispatched for emergency

42 SFS	Threat to personnel	Possibly shut down CVI/gates at discretion of operations superintendent/officer
42 SFS	Threat to equipment	Skywatch towers are brought down.
Vigilant Warrior	Threat to personnel.	Vacate open areas and take cover in hardened buildings
Lake Martin Rec Area	Threat to recreation area and campers	Vacate open areas and Lake Martin and take shelter in hardened buildings
LIGHTNING WATCH		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
All	Danger to life and property.	Be prepared to take shelter immediately upon issuance of the lightning warning
CAP-USAF	Potential to delay maintenance and refueling.	Be prepared to suspend maintenance and fueling operations.
Vigilant Warrior	Potential threat to personnel.	Be prepared to vacate open areas and take cover in hardened buildings
Lake Martin Rec Area	Potential threat to recreation area and campers	Be prepared to vacate open areas and Lake Martin and take shelter in hardened buildings
42 LRS/LGRF, 42 OSF/AOT	Potential threat to personnel.	Be prepared to discontinue all fueling ops.
42ABW/SFS, Fire Dept	Hazard to personnel.	Be prepared to advise personnel to seek shelter.
OBSERVED LIGHTNING WITHIN 5 NM WARNING		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
All	Danger to life and property.	Take shelter immediately.
CAP-USAF	Delays maintenance and refueling.	Suspend maintenance and refueling.
42 OSS/AOT	Delays aircraft recovery	Suspend aircraft recovery and fueling
AU	Cancels outdoor training	Direct students inside hardened buildings
42 LRS/LGRF, 42 OSS/AOT	Threat to personnel.	Discontinue all fueling ops.
42ABW Fire Dept	Hazard to flight line ops.	Advise personnel to seek shelter.
Cadet and Gunter Pools	Threat to swimmers	Vacate swimmers from the pools.
42 FSS/FSC	Hazard on Golf Course	Vacate the Golf Course
42 MSG/MPF	Hazard to computers	Backup computer systems
AU Library	Hazard to computers	Backup computer systems
NAF Computer Systems	Hazard to computers	Backup computer systems
42 CS / IT	Hazard to computers	Backup computer systems
42 SFS	Threat to personnel training	Direct students inside hardened facilities

42SFS	Threat to Military Working Dogs (MWDs)	Direct MWD Teams to hardened structures until dispatched for emergency.
42 SFS	Threat to response vehicles	Possibly parked under hardened area parking structure until dispatched for emergency
42 SFS	Threat to personnel	Possibly shut down CVI/gates at discretion of operations superintendent/officer
42 SFS	Threat to equipment	Skywatch towers are brought down.
Vigilant Warrior	Threat to personnel.	Take cover in hardened buildings
Lake Martin Rec Area	Threat to recreation area and campers	Vacate open areas and Lake Martin and take shelter in hardened buildings
FREEZING PRECIPITATION WARNING (ANY INTENSITY)		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
CAP-USAF	Ice on aircraft/runway.	Cancel missions.
42 OSS/AOT	1. Light: Icing on aircraft. 2. Any Icing: Runway and taxiways hazardous.	Hangar aircraft
AU	Cancels outdoor training	Direct students inside permanent buildings
42 SFS	Threat to personnel training	Direct students inside hardened facilities
42SFS	Threat to Military Working Dogs (MWDs)	Minimize utilization via coordination with the kennel master/trainer and at the discretion of operations superintendent/officer.
42 SFS	Threat to response vehicles	Park vehicles until needed for emergency dispatch
42 SFS	Threat to personnel	Possibly shut down CVI/gates at discretion of operations superintendent/officer
42 SFS	Threat to equipment	Skywatch towers are brought down.
Base Civil Engineer	1. Roadways dangerous. 2. Transformers may burst causing a loss of power and alarms may go off.	1. Plan alternate response routes. 2. Prioritize responses to alarms. 3. Decrease response speeds. 4. Sand roads.
HEAVY RAIN ≥ 2 INCHES IN 6 HOURS WARNING		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
CAP-USAF	Aircraft damage	Reschedule flight operations. Hangar aircraft.

	Runway/taxiway flooding	
42 CP	Lowland Flooding	Notify SFS/ECC of river levels
42 SFS	Lowland Flooding	Monitor / close floodgates
42 FSS/FSC	Golf Course Flooding	Close flooded holes
HEAVY SNOW ≥ 2 INCHES IN 12 HOURS WARNING		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
CAP-USAF	1. Ice/snow on aircraft 2. Runway condition changes	Cancel missions and/or hangar aircraft.
AU	Cancels outdoor training	Direct students inside for training.
42 OSS/AOT	Safety problems	May require divert to alternate bases without adequate maintenance support.
42SFS	Threat to Military Working Dogs (MWDs)	Minimize utilization via coordination with the kennel master/trainer and at the discretion of operations superintendent/officer.
42 SFS	Threat to personnel training	Direct students inside hardened facilities
42 SFS	Threat to response vehicles	Possibly parked under hardened area parking structure until dispatched for emergency Ensure snow tires or chains are installed on vehicles
42 SFS	Threat to personnel	Possibly shut down CVI/gates at discretion of operations superintendent/officer
Base Civil Engineer	Any Amount: Hazardous road/runway conditions ≥ 2 inches in 12 Hours Roadways dangerous	1. Plan alternate response routes 2. Decrease response speeds
BLIZZARD CONDITIONS (VISIBILITY ≤ 1/4 MILE WITH SNOW AND WIND ≥ 30 KNOTS) WARNING		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
CAP-USAF	Ice/snow on aircraft/runway	Cancel missions. Hangar aircraft.
AU	Cancels outdoor training	Direct students inside for training
42 OSS/AOT	Safety problems	May require divert to alternate bases without adequate maintenance support.
42SFS	1. May affect access to controlled and restricted areas 2. Abandoned/illegally parked vehicles may	1. Coordinate entry/exit procedures with the snow control center for snow removal. Ensure ropes and doughnuts are removed and replaced as needed. 2. Remove vehicles and control parking as conditions dictate.

	disrupt parking and traffic flow	
Base Civil Engineer	Any Amount: Hazardous road/runway conditions ≥ 2 inches in 12 Hours Roadways dangerous	1. Plan alternate response routes 2. Decrease response speeds
TEMPERATURE < 20 ºF ADVISORY		
CUSTOMER	**MISSION IMPACT**	**PROTECTIVE MEASURES**
ALL	Personnel in danger of frostbite	Advise personnel of the potential for frostbite
Base Civil Engineer	Increased unscheduled maintenance, damage to frozen pipes, heaters, buildings.	1. Reschedule routine work. 2. Repair pipes, buildings. 3. Restore heat
42 OSS/AOT	Equipment in danger of freezing	Secure Equipment in heated areas.
AU	Cancels outdoor training	Direct students inside for training
CAP – USAF WEATHER SENSITIVITIES (GA-8, C-172, and C-182)		
CRITERIA		**GO/NO-GO THRESHOLD**
Takeoff CIG / VIS		200 feet / ½ mile
Landing CIG / VIS / RVR		200 feet / ½ mile / 2400 ft RVR
Tailwind Component		Pilot discretion (normally 10 knots)
Crosswind		15 Knots
Winds Prevailing		30 Knots
Icing		No flying into areas of forecast or reported freezing rain or icing.
Turbulence		Flight operations are prohibited if severe turbulence (for CAT I aircraft) is known or reported. (Moderate for CAT II aircraft)
Lightning/Thunderstorms		Avoid thunderstorms/lightning by 5 NM.

Attachment 6

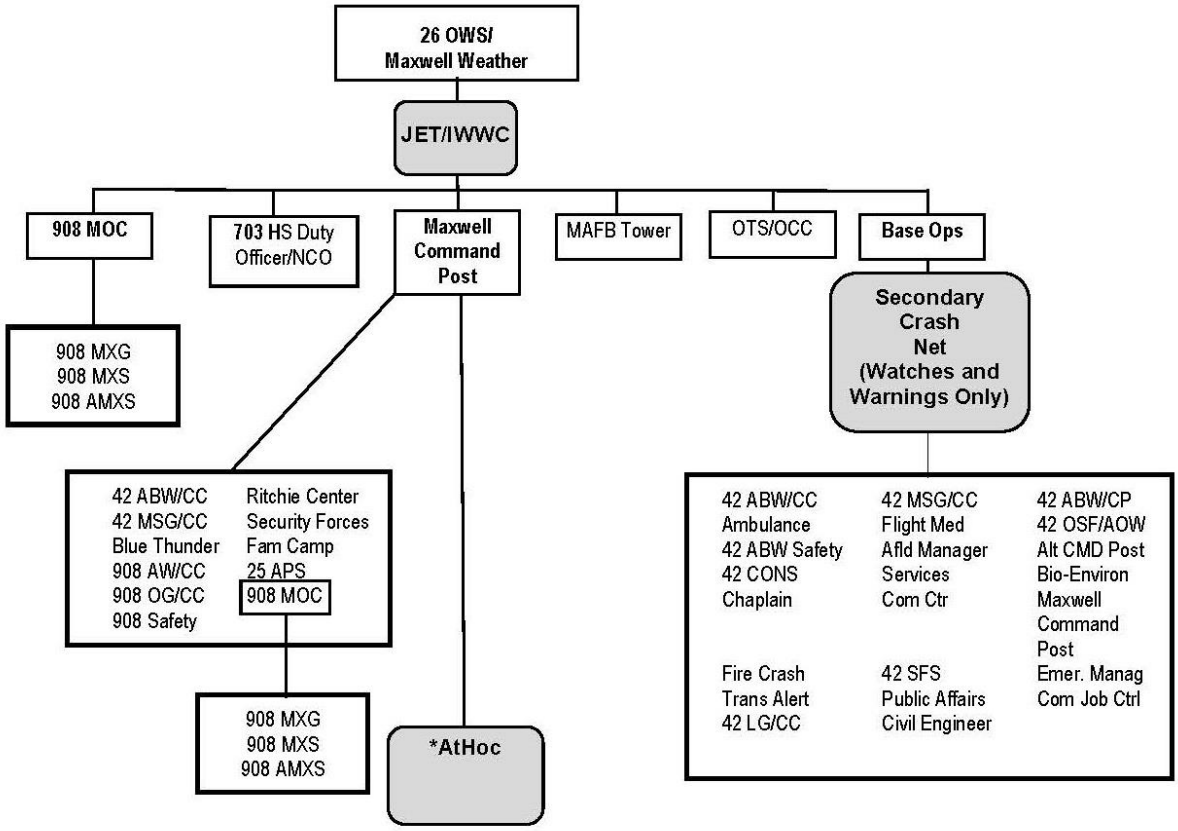
WEATHER WATCH/WARNING/ADVISORY NOTIFICATION CHAIN

A6.1. Resource Protection Notification Chain. Due to limited staffing and the time-critical nature of this information, 26 OWS personnel cannot individually notify every agency requiring weather watches, warnings, and advisories; hence, the application of a notification chain that exploits installation command and communications channels. Procedures developed to this end ensure weather personnel do not spend more time communicating than monitoring weather conditions. All units receiving these weather products must be involved in a continuous program of evaluation and improvement of the weather dissemination system, including inter-unit dissemination. Agencies must make certain that weather dissemination procedures ensure those needing information receive it. Individual commanders of units in need of weather information are responsible for having their units listed in the notification chain that follows.

Figure A6.1. 26 OWS to Maxell AFB and Lake Martin Notification Chain.

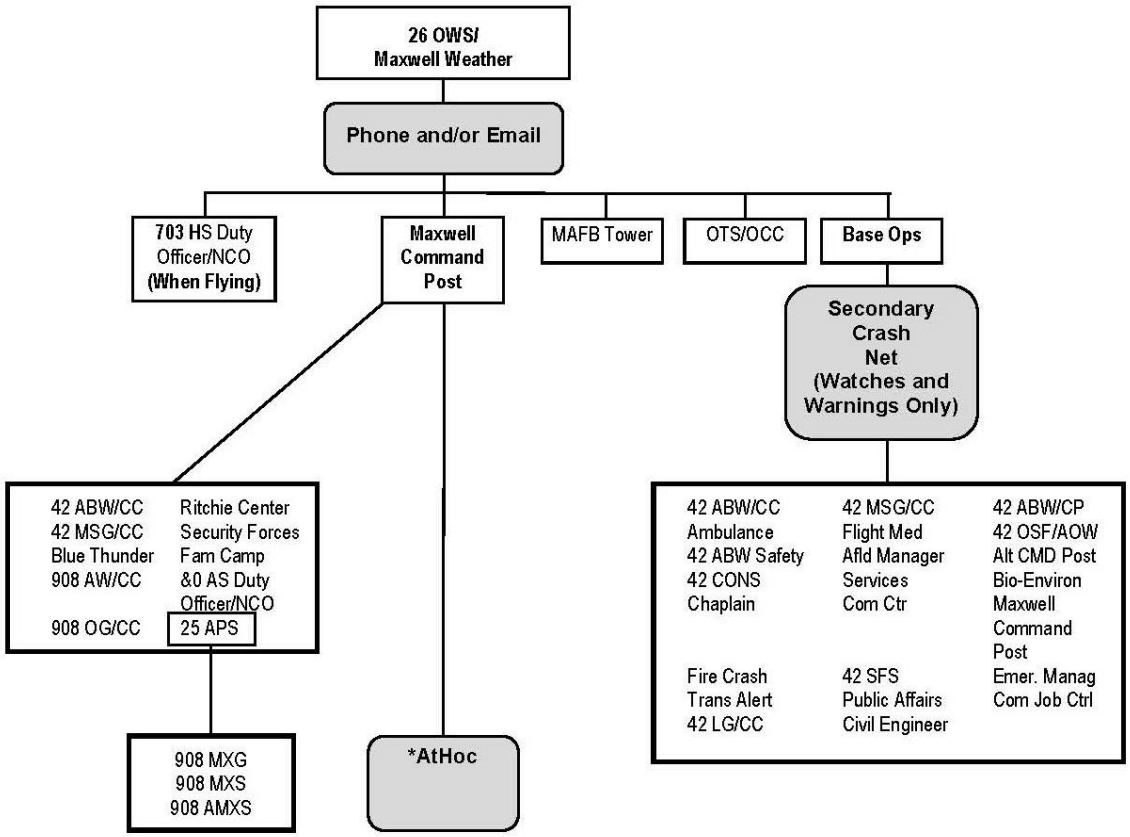


Figure A6.2. Weather Warning/Watch/Advisory Notification List.



* - AtHoc includes Giant Voice, automated phone calls, and computer pop-up notification.

Figure A6.3. Backup Weather Warninig/Watch/Advisory Notification List.



* - AtHoc includes Giant Voice, automated phone calls, and computer pop up notification.